

Healthy Choices, Lifecycle Health, and Earnings Inequality

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Abstract

How much can healthier choices reduce earnings inequality when mental and physical health both shape labor-market outcomes? We combine the United Kingdom Household Longitudinal Survey with a lifecycle model of employment and healthy choices—behavior and consumption—in which mental and physical health are separate margins and healthy choices are persistent. Persistence lets a moderate monetary cost of healthy consumption reproduce its income gradient. The model reproduces employment, healthy-choice, and health profiles by age and health state, and we use it to ask how much of the dispersion in cumulative earnings healthier choices could remove. Imposed on everyone, they reduce time spent outside good mental and physical health by 5.0% and the variance of log cumulative earnings between ages 25 and 55 by 13.4%. Mandates targeted at low offered earnings compress it further: under the P50 rule, the variance falls by 16.1%. How an intervention changes the resources that healthy choices require is central to its distributional effects.

Keywords: mental health; physical health; healthy choices; lifecycle earnings; inequality.

1 Introduction

Differences in health are among the most persistent sources of labor-market inequality over the life cycle. Yet health is not simply an exogenous risk that workers passively endure. Investments in physical activity, nutrition, and substance avoidance can alter health trajectories, while the time and resources available for these choices are themselves distributed unequally. This creates a cumulative feedback over the working life: poor health limits employment and earnings capacity, lower earnings constrain the ability to sustain healthy choices, and those choices help shape subsequent health. A sequence of modest differences can therefore cumulate into substantial disparities in lifetime earnings.

This paper asks how this feedback shapes earnings inequality and confronts a central policy tension. Are broad, population-wide improvements in healthy choices sufficient to compress earnings inequality, or does meaningful progressivity depend on reaching disadvantaged groups? Our object is therefore not simply the average health or earnings gain from healthier choices, but the distributional consequence of changing their incidence: how much would the dispersion of cumulative earnings change if healthy choices were more prevalent, and how would the answer depend on who changes them? Identifying where healthier choices

have the largest distributional effects can inform how policies are directed toward those who stand to benefit most, even when their broader objective is to improve population health.

Two distinctions are central to answering these questions. Health has multiple dimensions, among which mental and physical health are particularly relevant for economic outcomes. First, these dimensions need not move together or have interchangeable labor-market consequences. Treating health as a scalar obscures individuals who are in good health on one margin and poor health on the other, as well as the heterogeneity in labor-market penalties carried by different combinations of mental and physical health. Second, healthy choices use different resources. Physical activity primarily requires time, whereas another set of choices involving diet, smoking, and drinking requires goods expenditure. Income and employment can consequently affect the two choices through different channels. We retain both distinctions in the data and in the model.

Our empirical analysis uses the United Kingdom Household Longitudinal Survey (UKHLS), a large annual panel that follows individuals' health, labor-market outcomes, income, and health-related behaviors. Health is measured with the SF-12, a widely used twelve-item questionnaire that produces separate mental and physical component scores. The combination of these measures with repeated observations of employment, earnings, and behavior makes the UKHLS particularly well suited to our question: it allows us to follow the joint evolution of both health dimensions and economic outcomes, distinguish current health from persistent differences across people, and relate observed choices to subsequent health.

The separate SF-12 component scores allow us to preserve the distinction between mental and physical health. By crossing their good and poor classifications, we generate four health states: poor health on both dimensions, poor mental health only, poor physical health only, and good health on both. We also construct four permanent health types from persistent differences in individual health histories. Finally, we use principal-component analysis to summarize the set of observed healthy choices in the data as a vector of two indices, which we interpret as healthy behavior and healthy consumption.

Five facts motivate the quantitative analysis. First, mental and physical health follow different age profiles: average physical health declines substantially with age, while average mental health improves. Second, their labor-market associations are strongly non-additive. Relative to good health on both dimensions, employment is 3.8 percentage points lower with poor mental health alone and 7.4 points lower with poor physical health alone, but 31.5 points lower when both are poor. Third, both healthy-choice measures are less common below median income. Fourth, healthy choices are strongly persistent: 77.6% of individuals who made the healthy-consumption choice two years earlier make it again, against 23.6% of those who did not, and conditioning on the previous choice reduces the income gradient in healthy consumption from 8.1 to between 2.8 and 4.4 percentage points. Fifth, the two healthy choices are related differently to transitions in each health dimension. Both are associated with a higher probability of good physical health in the following wave, by 3.5 percentage points for healthy consumption and 2.8 points for healthy behavior. For mental health, only healthy consumption is associated with a higher probability of good health, by 1.8 points, whereas the estimate for healthy behavior is close to zero and statistically imprecise. These associations condition on lagged health and permanent health type.

We interpret these facts through an estimated finite-horizon model in which health, earn-

ings, and choices evolve jointly from ages 25 to 75. Individuals choose employment, saving, and two persistent healthy choices: healthy behavior and healthy consumption. Healthy behavior uses time, healthy consumption costs a fixed annual amount of money, and persistence arises because changing either choice relative to the previous year carries a utility cost. Current mental and physical health affect offered earnings, the utility cost of working, and survival, while the two healthy choices shift transition probabilities across the four future health states according to the conditional transition mapping estimated from the panel.¹ Permanent health type is correlated with permanent productivity, and a separate autoregressive shock generates earnings risk within a lifetime. The framework separates persistent sorting across people, changes in current health, employment selection, and innovations to earnings, rather than attributing the observed health–earnings gradient to a single mechanism.

We estimate health transitions, survival, the health-dependent earnings schedule, and the stochastic earnings process outside the structural optimization. The remaining preference and cost parameters and the distribution of permanent productivity conditional on health type are estimated using the method of simulated moments. The targeted moments describe employment and healthy-choice shares by age and health state, healthy-choice shares above and below median residual earnings, the persistence of healthy choices and the income gradient conditional on the previous choice, early-adult earnings means and variances by health type, and pooled early-adult earnings variances. Later working-life earnings dispersion and persistence provide validation beyond those initial-age targets. The model closely reproduces employment levels across health states, the principal healthy-choice profiles, the persistence of healthy consumption, and its income gradient, and it captures much of the earnings-dispersion profile after early adulthood. The estimated monetary cost of healthy consumption is about £2,900 a year.

We use quantitative model experiments to trace how prescribed healthier behavioral paths propagate through the estimated conditional health-transition process and employment into cumulative earnings. Under the universal benchmark, assigning both healthy choices to everyone reduces the share of alive person-years spent outside good mental and physical health by 5 percent and the variance of log cumulative earnings from ages 25 to 55 by 13.36 percent. We then construct fixed monetary thresholds at the tenth, twenty-fifth, and fiftieth percentiles of the baseline distribution of positive labor earnings, separately at each age, and assign both choices whenever offered earnings are at or below the relevant threshold. Under the median rule, the share outside good health falls by 3.46 percent and the variance of log cumulative earnings by 16.06 percent.

Additional experiments clarify the mechanism. Imposing both choices when mental and physical health are poor reduces the variance of log cumulative earnings by 9.12 percent, most of it through healthy consumption: a healthy-behavior mandate, whose only cost is time, improves health by a similar amount but compresses cumulative earnings by only 1.56 percent. Lowering the price of healthy consumption by 90 percent improves health modestly and raises dispersion by 2.02 percent, because it weakens the need to work most where earnings are lowest. When we instead remove individual health channels one at a time, variation in the

¹This mapping is an observational conditional expectation rather than an experimentally identified health-production function. We follow the direct panel approach used by Cole et al. (2019) and Mahler and Yum (2024), which estimate health transitions from observational data outside the structural estimation and carry the resulting schedules into their quantitative models.

utility cost of working is more important for earnings dispersion than variation in offered earnings, and mortality has a negligible effect in the common-survivor sample. Poor health therefore widens cumulative-earnings differences more through employment participation than through offered earnings conditional on work.

The paper contributes to the literature on health and labor supply by making the health process both multidimensional and responsive to individual choices. Existing lifecycle models quantify how health risk affects work, earnings, saving, and wealth (French, 2005; Capatina, 2015; Blundell et al., 2023; De Nardi et al., 2025; Hosseini et al., 2022). We retain their emphasis on persistent health heterogeneity, but allow mental and physical health to evolve separately and allow permanent heterogeneity to depend on both dimensions. This distinction is central because a scalar health stock cannot represent the different labor-market consequences of mixed health states or the severe, non-additive employment penalty associated with deficits in both dimensions.

The same multidimensional structure sharpens the health-investment mechanism. The health-capital literature, beginning with Grossman (1972), treats health as a stock that can be maintained through costly investment; subsequent work studies socioeconomic gradients in health behavior and embeds health effort or particular behaviors in quantitative lifecycle models (Cutler and Lleras-Muney, 2010; Galama and van Kippersluis, 2019; Cole et al., 2019; Darden, 2017; Mahler and Yum, 2024; Bueren et al., 2025). We extend this framework by distinguishing a time-intensive set of activities from an expenditure-intensive set of consumption choices, and by allowing both to persist through costs of adjustment, as Mahler and Yum (2024) do for effort. These inputs are not fungible: employment and income alter access to time and goods differently, and the two sets of choices can shift mental- and physical-health transitions by different amounts.

Tracing both inputs through joint health dynamics into cumulative earnings connects this mechanism to structural analyses of mental health and the labor market (Abramson et al., 2024; Ascarza-Mendoza and Velasquez, 2025; Jolivet and Postel-Vinay, 2025; Cozzi et al., 2024), evidence on mental illness, poverty, and earnings (Ridley et al., 2020; Biasi et al., 2021), and the literature on the origins of lifetime earnings inequality (Storesletten et al., 2004; Guvenen, 2009; Huggett et al., 2011). The contribution is therefore to show how asymmetric health risks and distinct investment resources interact with employment to shape earnings dispersion over the life cycle.

The paper proceeds as follows. Section 2 presents the lifecycle model. Section 3 describes the UKHLS and the construction of the health states, permanent types, and healthy-choice measures, and Section 4 documents the motivating empirical patterns. Section 5 explains the estimation strategy and evaluates model fit. Section 6 reports the counterfactual experiments, and Section 7 concludes. The appendix provides measurement details, the earnings-process identification, the health-transition recalibration, the numerical solution and weighting matrix, and complete counterfactual definitions.

2 Model

We develop a finite-horizon model in which health and earnings evolve jointly with employment, saving, and two persistent healthy choices. Health has two distinct but jointly evolving

dimensions in the model: mental health and physical health. Permanent heterogeneity and persistent earnings risk provide additional sources of differences across individuals.

2.1 Overview

Time is annual. Let $j = 1, \dots, J$, with chronological age $24 + j$ and $J = 51$. Individuals enter at age 25 and the terminal period is age 75. Conditional on being alive, the beginning-of-period state is

$$x_j = (a_j, z_j, H_j, F_j, \eta, \theta),$$

where a_j denotes assets, z_j persistent earnings risk, H_j current composite health, $F_j = (f_{j-1}^B, f_{j-1}^C)$ the pair of healthy choices made in the previous period, η permanent health type, and θ permanent log productivity. Both permanent characteristics remain fixed throughout life; current health and earnings risk evolve stochastically, and the previous choices evolve with the individual's decisions.

Each nonterminal period, the individual chooses a bundle $d_j = (n_j, f_j^B, f_j^C) \in \{0, 1\}^3$ and consumption c_j . Here n_j indicates employment, f_j^B healthy behavior, and f_j^C healthy consumption. Saving follows from the budget constraint. Changing either healthy choice relative to the previous period carries a utility cost, so the lagged choices F_j are part of the state. Bundle-specific taste shocks make discrete choice probabilistic conditional on the observed state. Health, earnings risk, and survival then determine the state in the following period, and the current healthy choices become next period's F_{j+1} .

The model is a partial-equilibrium environment with a fixed interest rate and an externally estimated schedule of offered earnings. It includes the extensive employment margin, but no endogenous hours or wage bargaining. The empirical hours results therefore motivate the relevance of health for labor-market outcomes without constituting a separate model prediction.

2.2 Health, healthy choices and survival

Write $H_j = (m_j, p_j)$, where m_j and p_j denote mental and physical health and each takes values Good or Bad. Throughout the paper, the first letter in a composite state refers to mental health and the second to physical health:

$$\mathcal{H} = \{BB, BG, GB, GG\}.$$

For example, BG denotes bad mental health and good physical health. A permanent health type, $\eta \in \{1, 2, 3, 4\}$, summarizes persistent differences in an individual's joint mental-physical health history. In the quantitative model, η is treated as an ex-ante endowment, fixed over the life cycle and known to the individual. Its empirical counterpart is a reduced-form classification intended to capture both permanent biological components of health and persistence in observed health trajectories. The types distinguish people whose health is persistently strong or weak in both dimensions from those whose long-run disadvantage is concentrated primarily in mental or physical health. Type shifts the transition probabilities, but it does not determine current health: individuals of the same type can occupy different states and move among them over time.

Conditional transitions satisfy

$$\Pr(H_{j+1} = h' \mid H_j = h, f_j^B, f_j^C, \eta) = \pi_j(h' \mid h, f_j^B, f_j^C, \eta). \quad (1)$$

The function π_j is the conditional transition mapping estimated from the panel. The counterfactual experiments impose policy invariance: when a choice pair (f_j^B, f_j^C) is assigned by an intervention, the model applies the same conditional transition probabilities estimated for that pair in the observational data.² The empirical starting point is a four-outcome multinomial logit containing current composite health, the two healthy choices, age, and permanent health type. We recalibrate the predicted probabilities so that the resulting transition system better reproduces one-year state retention and lifecycle health shares while preserving valid probability rows.³

Disciplining the effect of endogenous behavior on health transitions is a recurring challenge in this literature. Cole et al. (2019) estimate from PSID panel data a parametric transition function in which a composite of exercise and nonsmoking modifies a baseline transition matrix, restrict effort to favor health maintenance and improvement, and compare the implied effects with medical evidence. Mahler and Yum (2024) estimate a binary-health logit with current and lagged effort, education, current health, age, and a grouped fixed health type; their model also reproduces persistence in effort through adjustment costs. Both studies estimate the transition parameters directly from observational panels outside the structural estimation. Darden (2017) takes a different route by jointly estimating smoking choices, expectations, and health, thereby modeling the selection into behavior within the structural system. Bueren et al. (2025) instead infer persistent lifestyle types jointly from histories of multiple behaviors and health outcomes. Their health transitions depend on the latent lifestyle rather than on each contemporaneous behavior, shifting the object of interest from the short-run effect of a particular action to the long-run health trajectories associated with bundles of behavior.

Our approach is closest to the direct panel strategy of Cole et al. (2019) and Mahler and Yum (2024), and, like Mahler and Yum (2024), it generates persistence in healthy choices through costs of adjustment. It differs in what it asks the transition process to represent. We model movements among four joint mental-physical health states and allow two distinct contemporaneous choice margins to shift their distribution, rather than treating behavior as a single effort or a persistent lifestyle type. The choice coefficients are identified by variation in next-period health conditional on current health, age, and permanent type. The subsequent recalibration targets one-year state retention and lifecycle health shares, but does not provide additional identifying variation for the effects of the choices; likewise, the simulated-moments stage estimates preferences and costs taking the transition law as given. The policy-invariance condition stated above therefore remains a maintained restriction when this conditional expectation is used to generate counterfactual health histories.

Survival to the next period occurs with probability $s_j(H_j)$. Its direct dependence is on age and current health. Healthy choices can therefore affect subsequent survival through future

²This is a maintained structural assumption on the estimated conditional expectation, rather than a claim that π_j is an experimentally identified biological production function.

³Appendix C describes the two-step adjustment—first to origin-specific retention and then to the distribution of destination states—and reports the fitted recalibration parameters.

health, although they do not enter the current survival probability separately. Death has zero continuation value and there is no bequest motive. The constant flow b in equation (2) is received in every period alive. Because the value of death is normalized to zero, an additive constant in utility is no longer innocuous: b governs the value of an additional period of life and therefore the incentive to undertake choices that improve survival prospects.⁴

2.3 Preferences, earnings, and assets

The three discrete choices carry additive utility costs. For each healthy choice $q \in \{B, C\}$, the cost is quadratic in the model age index j and has state-specific intercepts relative to GG :

$$\psi_j^q(h) = \psi_0^q + \psi_1^q j + \psi_2^q j^2 + \Delta_h^q, \quad \Delta_{GG}^q = 0.$$

The employment cost is cubic in normalized age $t_j = (j - 1)/50$, with health-specific intercepts and polynomial interactions:

$$\psi_j^n(h) = \psi_0^n + \psi_1^n t_j + \psi_2^n t_j^2 + \psi_3^n t_j^3 + \Delta_h^n + \sum_{m=1}^3 \psi_{n,m}^h t_j^m, \quad \Delta_{GG}^n = 0, \quad \psi_{n,m}^{GG} = 0.$$

Allowing the entire schedule to differ by health state is important because employment participation has different levels and lifecycle profiles across BB , BG , GB , and GG . The flexible schedules let the model reproduce those differences while retaining the changes in participation incentives with age.

Healthy choices are also persistent. Changing a choice relative to the previous period carries a utility cost, κ_B for healthy behavior and κ_C for healthy consumption, whether the individual starts or stops the choice. In the data, healthy choices are strongly persistent: among individuals observed in consecutive behavioral modules, two years apart, 77.6% of those who made the healthy-consumption choice make it again, compared with 23.6% of those who did not; the corresponding shares for healthy behavior are 69.6% and 33.8%. The switching costs also change how permanent differences in resources affect healthy choices: adopting a choice becomes a commitment over several years rather than a decision renewed each period, so persistent differences in the cost of a choice relative to income weigh more heavily than transitory shocks.

Period utility, excluding the bundle-specific taste shock, is

$$u_j = \frac{(c_j^\alpha \ell_j^{1-\alpha})^{1-\gamma}}{1-\gamma} + b - \psi_j^n(H_j)n_j - \psi_j^B(H_j)f_j^B - \psi_j^C(H_j)f_j^C - \kappa_B \mathbf{1}\{f_j^B \neq f_{j-1}^B\} - \kappa_C \mathbf{1}\{f_j^C \neq f_{j-1}^C\}, \quad (2)$$

where leisure is $\ell_j = 1 - \phi_B f_j^B$ and $0 \leq \phi_B < 1$. Healthy behavior therefore uses discretionary time, whereas employment enters utility through its additive cost $\psi_j^n(H_j)$ rather than

⁴Its level is meaningful only relative to the units of consumption and the curvature of utility, so we calibrate it through a unit-free object: the value of a year of life expressed in years of consumption, $v = [u(\bar{c}) + b]/[u'(\bar{c})\bar{c}]$, evaluated at mean simulated consumption. The static calibration in Hall and Jones (2004) implies $v \approx 4$; we set $b = 0.011$, which delivers the same value in our model at $\gamma = 2$.

subtracting hours directly from the time endowment. Under this specification, employment and healthy behavior do not compete within a physical time constraint; their interaction operates through utility, income, saving, and the monetary cost of healthy consumption.⁵

Offered annual labor earnings are

$$w_j = \exp\{g(24 + j) + \lambda_{H_j} + \theta + z_j\}, \quad z_{j+1} = \rho z_j + \nu_{j+1}, \quad \nu_{j+1} \sim N(0, \sigma_\nu^2), \quad (3)$$

where g is the predictable lifecycle profile, λ_h is the health-state earnings differential relative to GG , and θ is permanent productivity. The functions g and λ_h are estimated from the earnings of employed individuals and enter the model's offered-earnings schedule. Employment is chosen endogenously, and realized labor income is $y_j = n_j w_j$. Permanent log productivity follows

$$\theta \mid \eta = k \sim N(\mu_{\theta,k}, \sigma_{\theta,k}^2). \quad (4)$$

Consequently, health and productivity can be associated at entry without requiring current health to explain every earnings difference. Because the baseline initializes $z_1 = 0$, the conditional distribution of θ captures the initial earnings variance not accounted for by current health and the deterministic age profile. Persistent innovations then generate additional heterogeneity after age 25.

The budget constraint is

$$c_j + a_{j+1} + p_f f_j^C = (1 + r)a_j + n_j w_j, \quad a_{j+1} \geq \underline{a}. \quad (5)$$

Healthy consumption has a monetary cost p_f in addition to its direct utility cost: an annual expenditure, in 2015 pounds, paid in every period in which $f_j^C = 1$, whether or not the individual works. A common monetary amount weighs more heavily on individuals with lower resources, which is how the model relates the prevalence of healthy consumption to income. Initial assets are fixed at £21,085, the mean net wealth observed among individuals aged 20–24. Because the model begins at age 25 and the wealth data are available in five-year age groups, we use this value as an estimate of net wealth at entry. Individuals cannot borrow, so total resources must cover consumption and the cost of healthy consumption in every period. Saving provides a way to transfer resources across periods, but does not insure permanent productivity or health risk fully. The model has no public transfers or non-labor income, so an individual without assets finances consumption and any healthy-consumption expenditure through work.

2.4 The individual's problem

Let $v_j(x, d)$ denote the value of bundle $d = (n, f^B, f^C)$ before its taste shock. For $j < J$,

$$v_j(x, d) = \max_{c, a'} \{u_j(c, d, H, F) + \beta s_j(H) \mathbb{E}[V_{j+1}(a', z', H', F', \eta, \theta) \mid x, d]\}, \quad F' = (f^B, f^C), \quad (6)$$

⁵We also estimated a specification in which employment used time. Its health-specific time costs could not be separately identified from the additive employment costs with the available participation and behavior variation. Appendix D presents that specification and the evidence behind our choice.

subject to the budget and borrowing constraints. The expectation uses the earnings process and the choice-dependent health transition law; the next-period lagged choices are the choices made today. Independent type-I extreme-value taste shocks, with scale τ , imply

$$P_j(d \mid x) = \frac{\exp\{v_j(x, d)/\tau\}}{\sum_{\tilde{d}} \exp\{v_j(x, \tilde{d})/\tau\}}, \quad V_j(x) = \tau \log \sum_d \exp\{v_j(x, d)/\tau\}, \quad (7)$$

under the location normalization used in the implementation. Because the switching costs enter utility additively, the optimal consumption and saving policy for a given bundle does not depend on F ; the lagged choices affect which bundle is chosen.

At age 75 there is no continuation value and the individual consumes all remaining resources. Employment nevertheless remains a choice: its contemporaneous earnings are weighed against the utility cost of working. In the unrestricted problem, both healthy choices are set to zero by terminal convention because their modeled health benefit operates exclusively through next period’s transition, which lies beyond the horizon; the terminal convention carries no switching cost. Terminal choices must satisfy the solvency condition $a_{J+1} = 0$ with $c_J > 0$. This restriction is propagated backward through the Bellman problem, so saving choices at earlier ages must leave sufficient resources for every subsequent state reached with positive probability, including the expenditure on healthy consumption. It therefore imposes no arbitrary age profile on saving and requires no terminal consumption floor. Imposed-choice counterfactuals maintain their stated restrictions through the terminal period, although terminal choices cannot affect a subsequent health transition. Because behavior near this boundary is shaped by the finite horizon, the earnings-distribution validation focuses on ages 25–60 and the cumulative-earnings counterfactuals on ages 25–55.

3 Data

The analysis uses the first twelve waves of the UKHLS, fielded between 2009 and 2022. Repeated individual observations connect health, labor-market outcomes, income, and behaviors.⁶ The raw merged panel contains 933,102 person-years from 128,017 individuals. Requiring non-missing mental and physical SF-12 scores, complete age records, and positive longitudinal survey weights reduces the sample to 207,738 person-years from 36,794 individuals. Most of this initial reduction is associated with eligibility for longitudinal weights: these weights require prior survey participation and are not defined in wave 1, so the estimation panel begins in wave 2.

We then apply the labor-market data requirements, trim the top one percent of real hourly wages, restrict the sample to ages 25–76, and remove duplicate person-age records and two remaining log-wage outliers. The resulting estimation sample contains 140,879 person-years from 26,853 individuals, observed for 5.2 waves on average.⁷ Descriptive statistics and empirical moments use the UKHLS longitudinal weights, which are designed to adjust for differential non-response and panel attrition along observable characteristics. Appendix

⁶The panel structure allows us to distinguish persistent health differences from current states and to study earnings covariances at different lags.

⁷This is an unbalanced panel by design: respondents enter the usable sample in different waves, longitudinal weights require prior participation, and not every module is fielded each year.

Table A.1 reports the complete sequence of filters, and Appendix A.3 examines differential exit from the panel.

An important advantage of the UKHLS is that, in addition to rich demographics, labor-market outcomes, income, and annual health measures, it records several health-related behaviors. We observe vigorous and moderate physical activity, walking, fruit and vegetable consumption, smoking, and heavy drinking. These questions are distributed across periodic modules: physical activity and alcohol are available in waves 7, 9, 11, and 12; fruit and vegetable consumption in waves 2, 5, 7, 9, and 11; and smoking in waves 6 through 12. All seven inputs overlap in waves 7, 9, and 11, yielding 24,214 person-years from 11,282 individuals for the two healthy-choice measures. Choice shares, their income gradients, and the health transition estimations use this behavioral subsample, whereas health, employment, and earnings profiles use the broader panel. Appendix A elaborates on sample selection. Table 1 summarizes the estimation panel by composite health state.

Table 1: Descriptive statistics by composite health state

	GG	GB	BG	BB
<i>Demographics</i>				
Age (years)	47.3 (14.1)	55.8 (14.1)	42.6 (12.5)	51.9 (13.8)
Female (%)	47.8 (50.0)	51.9 (50.0)	55.4 (49.7)	58.9 (49.2)
Higher education (%)	47.2 (49.9)	32.8 (46.9)	47.8 (50.0)	26.8 (44.3)
<i>Labor-market outcomes</i>				
Employed (%)	69.9 (45.9)	46.1 (49.8)	73.9 (43.9)	35.8 (47.9)
Annual labor earnings (2015 £)	15364 (13404)	8943 (11580)	15288 (12471)	6412 (9940)
Weekly hours, employed	35.0 (9.0)	33.8 (9.5)	34.4 (8.8)	33.3 (9.4)
Hourly wage, employed (2015 £)	12.74 (5.39)	11.54 (4.86)	12.22 (5.11)	10.80 (4.38)
<i>Healthy choices</i>				
Healthy behavior, $f^B = 1$ (%)	49.9 (50.0)	35.6 (47.9)	44.9 (49.7)	30.0 (45.8)
Healthy consumption, $f^C = 1$ (%)	64.5 (47.8)	60.6 (48.9)	53.8 (49.9)	45.5 (49.8)
Person-year observations	52,315	34,279	28,062	26,223

Notes: Each cell reports a weighted mean and standard deviation in parentheses from the UKHLS estimation panel. The state label lists mental health first and physical health second. Earnings include zero earnings for people not employed; hours and wages are conditional on employment. The healthy-choice rows use the person-years in which all behavioral inputs are observed.

3.1 Sample selection

The data preparation retains age 76 only as an auxiliary next-wave observation for estimating health transitions at the upper boundary; all model outcomes and moments end at age 75. Employed observations with missing labor earnings or hours are excluded from the wage sample.

Monetary variables are deflated to 2015 pounds with the UK Consumer Prices Index for all items. Annual labor earnings are constructed by annualizing observed monthly real labor income. A real hourly wage is constructed as real monthly labor earnings divided by four times usual weekly hours. Three filters then apply to employed observations: weekly hours

must lie between 10 and 100; real hourly wages must be at least £5.72 in 2015 pounds; and the top one percent of the real hourly-wage distribution among the employed is trimmed.⁸ Labor-outcome regressions and the external stochastic earnings-process estimation use working ages 25–60. Appendix A records additional implementation details.

3.2 Measuring mental and physical health

Mental and physical health are measured with the SF-12, a twelve-question instrument derived from the SF-36 that summarizes self-reported health in separate Mental Component Summary (MCS) and Physical Component Summary (PCS) scores. Higher values indicate better health on the corresponding scale. The instrument was developed and validated by Ware et al. (1996), and its ability to distinguish mental-health conditions has also motivated its use with the UKHLS by Jolivet and Postel-Vinay (2025). We retain the continuous scores when describing age profiles.

To introduce the corresponding health measures into the model parsimoniously, we discretize each continuous SF-12 score: the bottom two weighted quintiles are classified as Bad and the top three as Good. The thresholds are guided by regressions of employment and log earnings on score-quintile indicators, reported in Appendix Table A.4. For physical health, the PCS is a useful criterion because it summarizes limitations in physical functioning on a common scale, and the regressions show markedly larger employment and conditional-earnings penalties in its bottom two quintiles than toward the top of the distribution. For mental health, the threshold is close to the one in Jolivet and Postel-Vinay (2025), whose Poor and Severe categories together cover roughly the bottom 36 percent of the MCS distribution. Crossing the two classifications produces *BB*, *BG*, *GB*, and *GG*, with mental health listed first. This distinction separates poor physical functioning with good mental health from poor mental health with good physical functioning, which a single indicator for any poor health would obscure.

3.2.1 Permanent health types

Permanent health types capture persistent individual differences across the joint distribution of mental and physical health. We residualize the continuous mental and physical summary scores on a quadratic polynomial in age and indicators for sex and education. We then average each residual within individual across all available panel waves and partition the resulting two-dimensional individual means into four clusters using *k*-means. The resulting labels describe four distinct profiles: relatively good health in both dimensions, poor health in both, predominantly poor physical health, and predominantly poor mental health.

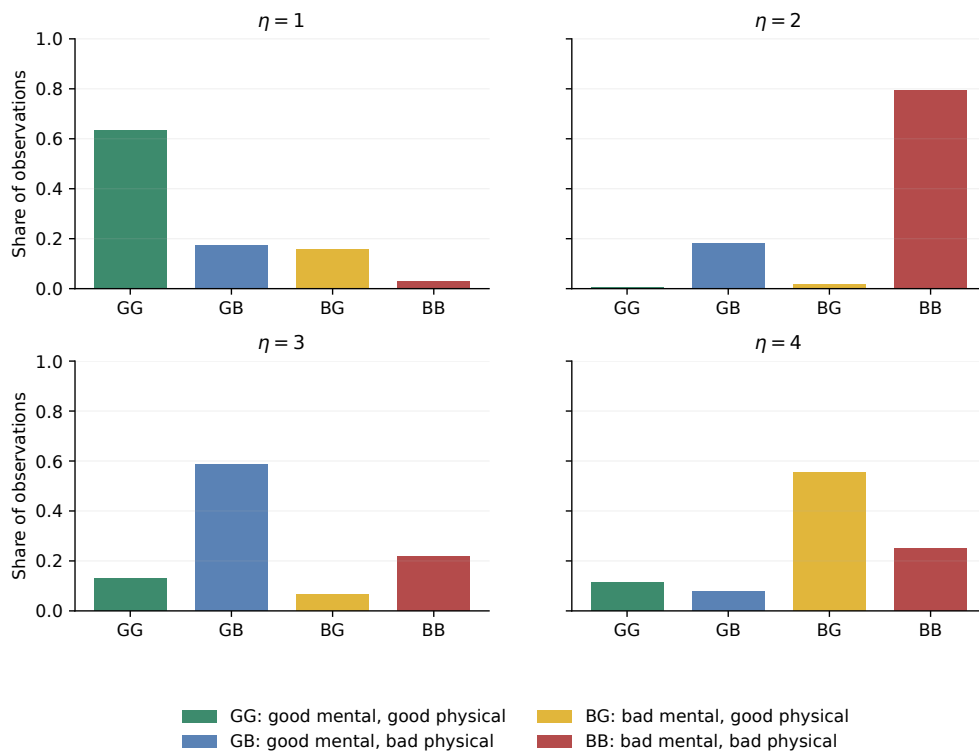
Types are fixed over time, whereas composite health states can change each year. In the quantitative model, type is represented as a fixed latent endowment known to the individual at entry. Empirically, using all of an individual’s observed waves helps separate persistent heterogeneity from transitory variation in the SF-12 scores. The resulting classification should therefore be understood as a reduced-form summary of long-run health persistence,

⁸Applying both wage restrictions to the real measure keeps the lower cutoff and the pooled upper-tail trim comparable across survey years rather than allowing inflation to change their effective incidence.

reflecting both permanent biological components and persistence in realized health trajectories, rather than as a baseline characteristic measured exclusively before age 25. Appendix A provides further construction details.

Figure 1 shows that each permanent type is associated primarily with the corresponding current health state. Type 1 spends 63.6% of its observed person-years in *GG*, while type 2 spends 79.5% in *BB*. The predominant state is *GB* for type 3, accounting for 58.6% of its observations, and *BG* for type 4, accounting for 55.3%. The mapping is nevertheless not deterministic: individuals occupy other current states within every type. This combination of a clear modal state and meaningful within-type variation motivates including both permanent type and current health in the transition process.

Figure 1: Composite health states by permanent health type



Notes: Weighted share of person-year observations in each composite health state, ages 25–75. The first state letter denotes mental health and the second physical health.

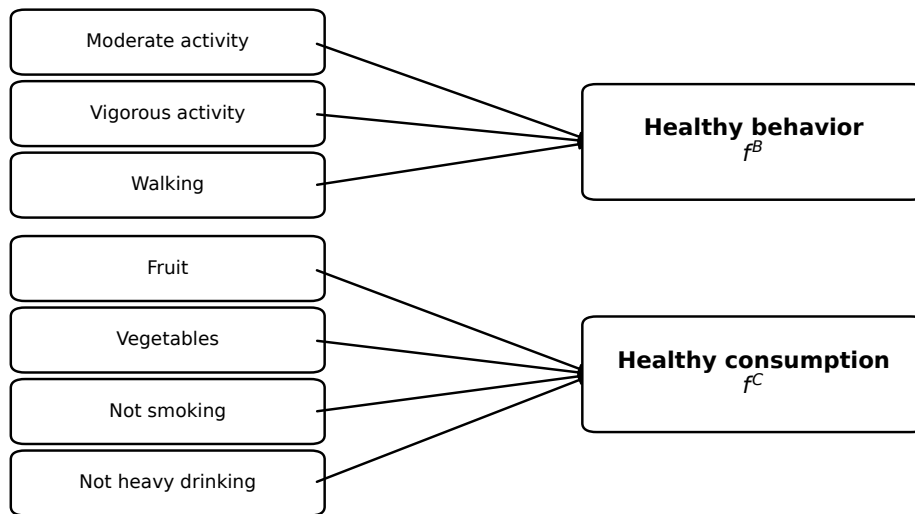
3.3 Measuring healthy choices

The structural model requires a parsimonious representation of the health-related behaviors observed in the data. We collected seven variables from the data—vigorous activity, moderate activity, walking, fruit consumption, vegetable consumption, smoking, and heavy drinking—which we reduce to two dimensions using survey-weighted principal components followed by a varimax rotation. Two components have eigenvalues above one, 1.75 and 1.41 against 0.92 for the third, and together account for 45% of the total variance. One component loads primarily on vigorous activity, moderate activity, and walking, and we

label it healthy behavior. The other loads on fruit and vegetable consumption, not smoking, and not drinking heavily, and we label it healthy consumption. The latter is therefore a consumption-related behavior bundle, not a direct observation of medical spending or a single food purchase. Appendix Table A.5 reports the rotated loadings.

To construct f^B and f^C , we split each rotated component score at zero. This cutoff is the reference point induced by centering the variables used in the weighted PCA and therefore defines a sample-based relative partition. A positive score means that the individual's standardized behavior bundle projects in the healthy direction relative to that reference distribution, while a negative score places it on the opposite side. After applying the estimation-sample filters, the mean scores are -0.06 for behavior and 0.20 for consumption. The zero cutoff classifies 40% of person-years as engaging in healthy behavior and 65% as engaging in healthy consumption. Figure 2 summarizes the construction. The reduction makes the decision problem tractable but removes within-category intensity differences, so an intervention setting $f^B = 1$ represents a switch in the model's binary behavior category, not a specified increase in minutes of exercise.

Figure 2: Measurement of healthy choices



Notes: Seven observed behaviors grouped into healthy behavior (f^B) and healthy consumption (f^C).

4 Facts about healthy choices, lifecycle health and earnings

Five empirical patterns motivate the model. First, mental and physical health follow divergent lifecycle profiles: physical health deteriorates while mental health improves with age. Second, joint deficits in mental and physical health are associated with severe labor-market penalties that a physical-health measure alone would obscure. Third, healthy behavior and

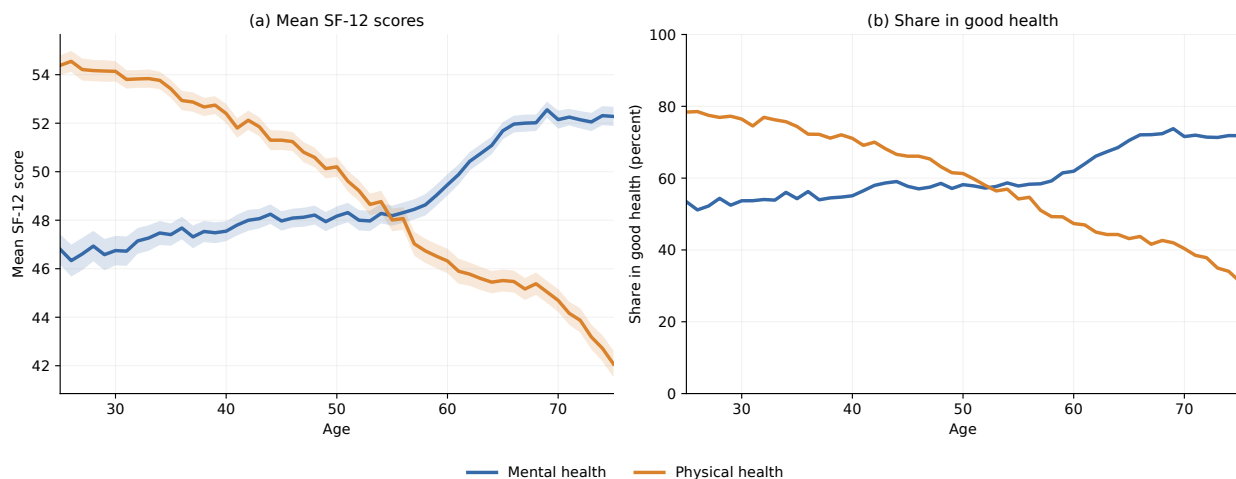
healthy consumption are less prevalent among lower-income individuals. Fourth, healthy choices are strongly persistent, and much of their income gradient is carried by choices made in the past. Fifth, after conditioning on lagged health and permanent health type, lagged healthy choices remain positively associated with subsequent physical health, whereas their predictive associations with subsequent mental health are weaker. We document these associations and their uncertainty before imposing the dynamic structure used in the counterfactual exercises.

4.1 Lifecycle health and healthy choices

Figure 3 describes how both the continuous SF-12 measures and their discrete model counterparts evolve over the lifecycle. Panel (a) shows that the two health dimensions follow markedly different paths. The mean physical score declines from 54.4 at age 25 to 46.3 at age 60 and 42.1 at age 75, while the mean mental score rises from 46.8 to 49.5 and 52.3 at the same ages. The divergence becomes especially pronounced later in the working life, when the deterioration in physical functioning accelerates as mental health continues to improve.

Panel (b) shows that discretization preserves this contrast. The share in good physical health falls from 78.4% at age 25 to 47.3% at age 60 and 31.2% at age 75. Over the same ages, the share in good mental health increases from 53.4% to 61.9% and 71.8%. Good mental health becomes more prevalent than good physical health from the mid-fifties onward. The model must therefore accommodate not only age-related health change, but also the opposing evolution of its mental and physical dimensions.

Figure 3: Mental and physical health over the life cycle

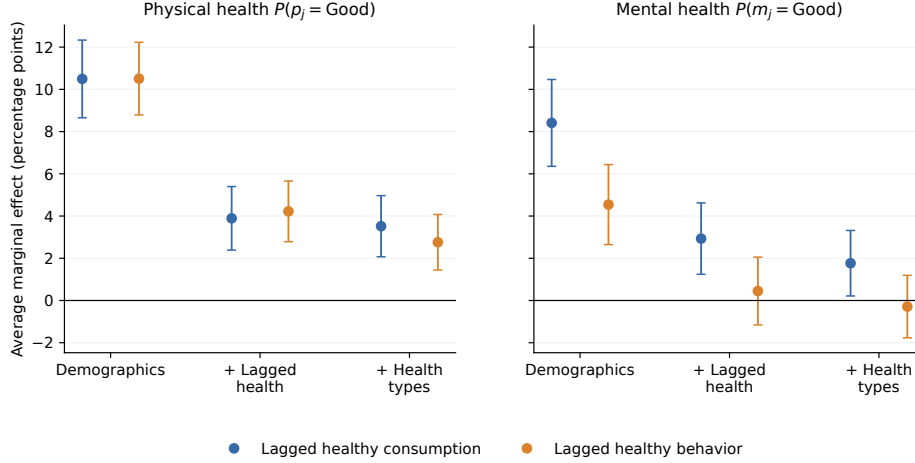


Notes: Panel (a) reports weighted mean SF-12 mental and physical health scores; shaded bands are pointwise 95% confidence intervals clustered by individual. Panel (b) reports the weighted share classified in good health in each dimension.

Figure 4 reports separate logits for good physical and good mental health. All specifications include both lagged healthy choices. Within each specification, the pair of points reports the average discrete marginal effects of lagged healthy consumption and lagged healthy behavior, respectively. The left panel uses good physical health as the outcome and the right

panel uses good mental health. The first specification includes a quadratic in age and indicators for year and sex, the second adds separate indicators for lagged good mental and physical health, and the third also includes permanent health types. The regressions are weighted by the longitudinal survey weight; the points are expressed in percentage points and the whiskers are individual-clustered 95% confidence intervals. All six regressions use a common consecutive-wave sample of 19,566 observations from 9,613 individuals.

Figure 4: Lagged healthy choices and subsequent physical and mental health



Notes: Average marginal effects of lagged healthy choices on the probability of good physical and good mental health, in percentage points, from weighted logits. Whiskers are individual-clustered 95% confidence intervals.

Adding health controls substantially attenuates the associations. For good physical health, the marginal effects of consumption and behavior fall from 10.49 and 10.51 percentage points with demographic controls alone to 3.52 and 2.76 points in the full specification. Their full-specification 95% intervals are [2.07, 4.97] and [1.44, 4.07] points. For good mental health, the corresponding full-specification effects are 1.77 and -0.29 points; the interval for consumption excludes zero narrowly, whereas the interval for behavior includes zero. Thus, this figure provides stronger conditional evidence for the physical-health association than for the mental-health association.

4.2 Why mental health is important for earnings inequality

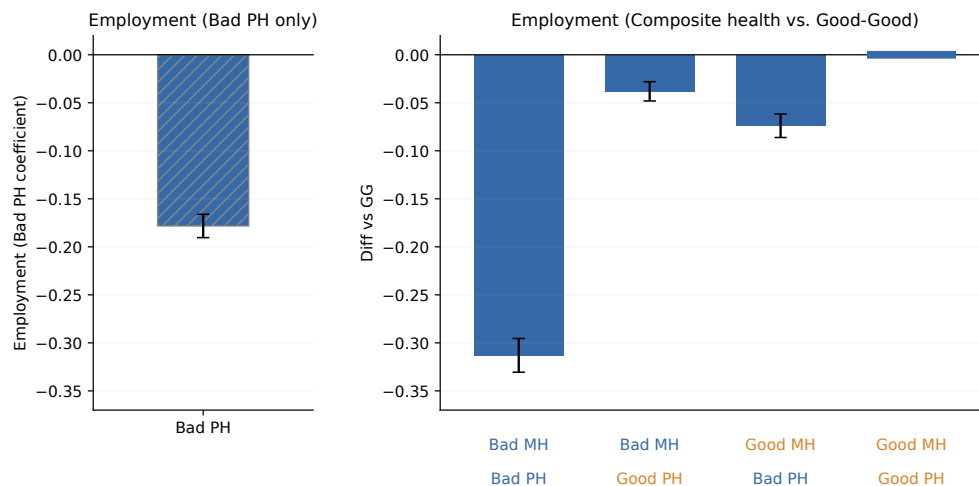
Representing health only through physical limitations can conceal economically important heterogeneity. Two individuals with the same physical-health classification may differ in mental health and, consequently, in their attachment to employment or their earnings capacity. A physical-health indicator pools these individuals and attributes to one dimension differences that may arise from either one. This limitation also applies to lifecycle models that introduce health exclusively through physical functioning: they cannot distinguish workers in *BB* from those in *GB*, or workers in *BG* from those in *GG*.

We assess the importance of this distinction in two steps. In both, *GG* is the reference state. The regressions control for a quadratic in age and indicators for year, sex, race, marital

status, urban residence, and education; they use the longitudinal survey weight and cluster standard errors by individual.

First, Figure 5 shows what is lost by using physical health alone. It compares an employment regression containing a single indicator for bad physical health with a specification that replaces it with the four composite mental–physical health states. The bad-physical-health indicator is associated with a 17.83 percentage-point employment gap. Once mental health is distinguished, however, the employment gap is 31.30 points for *BB* and 7.39 points for *GB*. Thus, among individuals classified in bad physical health, the two mental-health states are separated by almost 24 percentage points in employment. The composite measure also identifies a 3.81-point employment gap for *BG*, which a physical-health-only specification assigns to the good-health reference group.

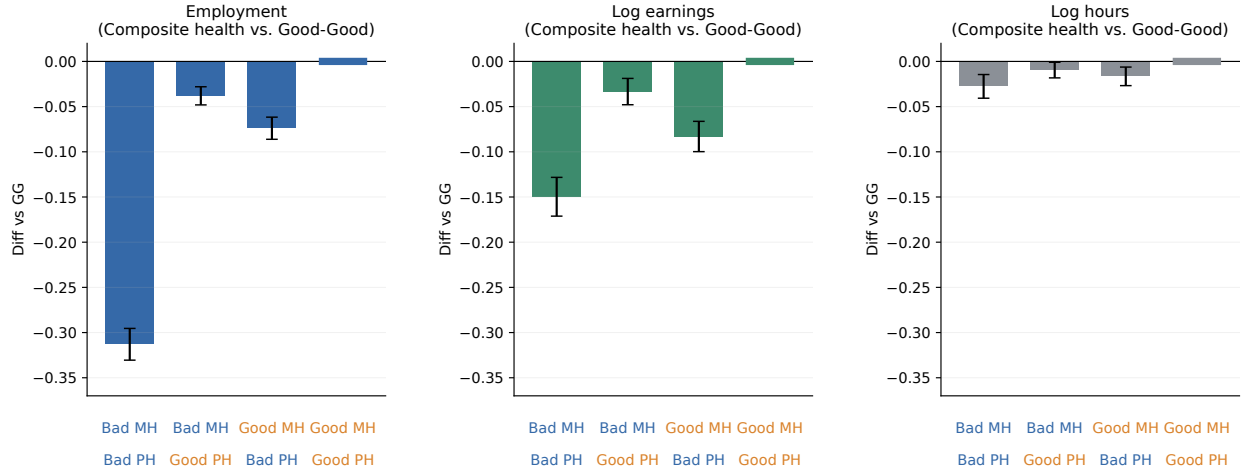
Figure 5: Employment: bad physical health versus composite health states



Notes: Employment regression coefficients for bad physical health alone and for composite health states relative to good mental and physical health (GG), ages 25–60. GG is displayed at its normalized reference value of zero; whiskers for estimated coefficients show individual-clustered 95% confidence intervals.

Second, Figure 6 uses the composite states to examine where the additional heterogeneity appears: employment participation, log annual earnings conditional on employment, or log weekly hours conditional on employment. The penalties are most pronounced in employment and earnings rather than in weekly hours among those who remain employed. Relative to *GG*, *BB* is associated with a 31.30 percentage-point reduction in employment and a 0.150-log-point reduction in conditional annual earnings. The corresponding estimates are 7.39 percentage points and 0.083 log points for *GB*, and 3.81 percentage points and 0.033 log points for *BG*. Conditional-hours differences are smaller: 0.028 log points for *BB*, 0.017 for *GB*, and 0.010 for *BG*.

Figure 6: Labor-market outcomes by composite health state



Notes: Employment, log earnings conditional on employment, and log hours by composite health state, relative to GG, ages 25–60. All panels use a common vertical scale; coefficients retain the units of their respective outcomes. GG is displayed at its normalized reference value of zero, and whiskers for estimated coefficients show individual-clustered 95% confidence intervals.

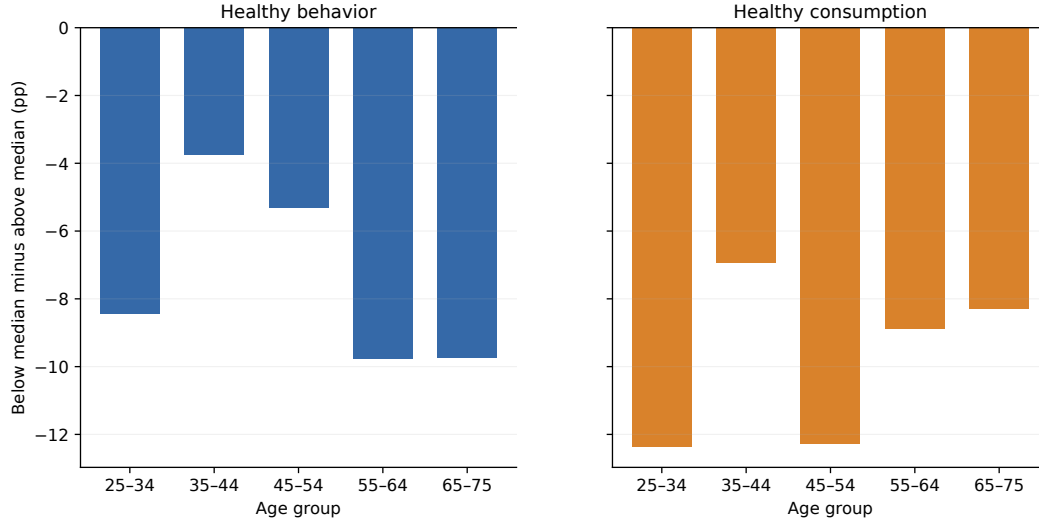
These patterns guide the model specification. Composite health affects both the employment decision and offered earnings, allowing mental and physical health to generate heterogeneity along the two margins that are most prominent in the data. Given the empirical dominance of the participation margin and the added complexity of modeling hours jointly with employment, the model abstracts from an intensive hours choice. This is a parsimonious modeling choice rather than a restriction implied directly by the conditional-hours estimates. The model accommodates part of this heterogeneity through the joint distribution of permanent health type and productivity.

4.3 Earnings and healthy choices

This subsection documents how healthy choices vary with income. We first compare the prevalence of each choice above and below median income in the raw data. We then hold age and health fixed and compare individuals with different earnings, which is the comparison the estimation uses to discipline the monetary cost of healthy consumption.

Figure 7 compares the share making each healthy choice below and above the pooled weighted median of real gross monthly income. The median is £1,640 in 2015 prices, and the bars report below-minus-above differences. Healthy behavior is less common in the lower-income group by 4.2–9.8 percentage points across age groups; healthy consumption is less common by 6.9–12.3 points. Both differences remain present in the 65–75 group.

Figure 7: Healthy choices below versus above median income



Notes: Difference in the weighted share making each healthy choice below versus above the pooled median of real gross monthly income, by age group. The income median is computed over person-year observations aged 25–75; bars show percentage-point differences.

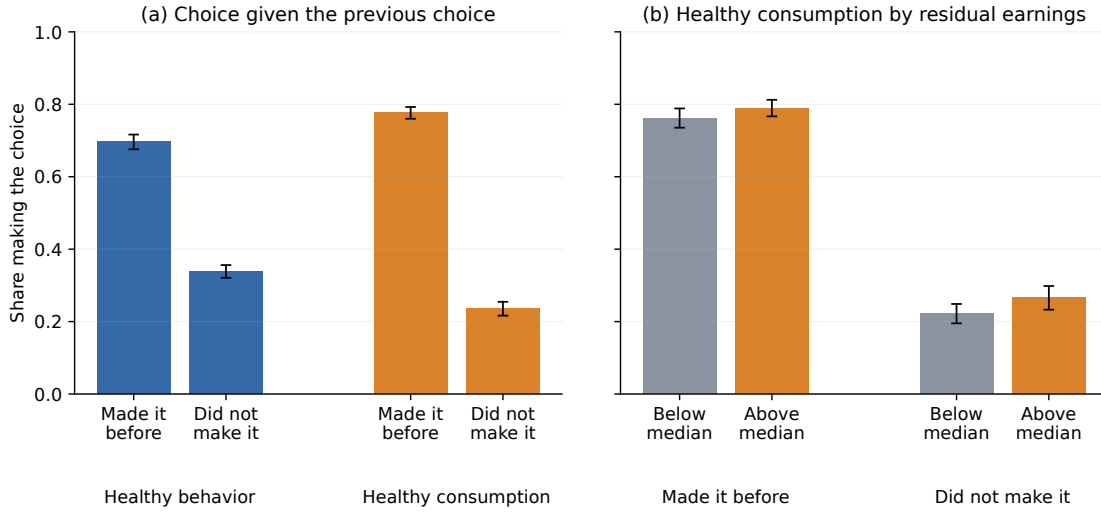
The raw comparison in Figure 7 mixes two forces. Lower-income person-years differ from higher-income ones in age and health, and age and health affect healthy choices directly, independently of income; in the model, the utility costs of both choices vary with age and health state. A raw income gap could therefore reflect who is in the lower-income group rather than the difference in income itself. To isolate the role of earnings, we compare individuals of similar age and health who differ in earnings. Among person-years with positive labor earnings at ages 25–60, we regress log labor earnings on a quadratic in age and indicators for composite health, and split the sample at the weighted median of the residual. Because the residual is uncorrelated with age and health, the two halves have approximately the same age and health composition, and differences in healthy choices between them reflect differences in earnings among comparable individuals. Holding age and health fixed in this way, the healthy-behavior gap falls to 0.45 percentage points, whereas a healthy-consumption gap of 8.09 points remains: the income gradient in healthy consumption is not explained by age or health composition. These residualized gaps are the moments used in estimation. Because they also use labor earnings rather than total income and a narrower age range, they need not reproduce the bars in Figure 7.

4.4 Persistence of healthy choices

Healthy choices are not renewed independently each year. Because the behavioral module is fielded every other wave, we compare an individual’s choice with the same individual’s choice at the previous observation of the module, one to three years earlier and almost always two. Figure 8(a) shows that the choices are strongly persistent. Among individuals aged 25–60 who made the healthy-consumption choice at the previous observation, 77.6% make it again, compared with 23.6% of those who did not; the difference is 54 percentage points. For healthy behavior the corresponding shares are 69.6% and 33.8%. Persistence of

this magnitude is consistent with habits, costs of changing established routines, or persistent unobserved preferences, and the cross-sectional choice shares do not distinguish among them.

Figure 8: Persistence of healthy choices



Notes: Ages 25–60. Panel (a) reports the weighted share making each choice among individuals who did and did not make it at their previous observation of the behavioral module, one to three years earlier. Panel (b) reports the healthy-consumption share below and above the weighted median of residual log labor earnings (age, age squared, and composite health removed), by the previous healthy-consumption choice. Whiskers are individual-clustered 95% confidence intervals.

Persistence also shapes the income gradient. Panel (b) splits the individuals in panel (a) at the median of residual labor earnings used above. Conditional on the previous healthy-consumption choice, the income gradient narrows substantially: 2.8 percentage points among those who made the choice before (79.0% above the median against 76.2% below) and 4.4 points among those who did not (26.6% against 22.2%), compared with 8.1 points without conditioning. Much of the income gradient therefore reflects who was already making healthy choices, rather than a response of current choices to current resources. The gradient is also largest early in working life: at ages 25–29, 51.7% of the upper half and 38.3% of the lower half make the healthy-consumption choice. These facts motivate the costs of changing choices in the model and the moments used to estimate them, and they discipline how much of the income gradient the monetary price of healthy consumption must explain.

5 Estimation

We estimate the model using a two-step approach following Gourinchas and Parker (2002). The first step estimates or calibrates the earnings process, health transitions, survival, and initial conditions outside the dynamic optimization problem. Taking these inputs as given, the second step estimates preference and cost parameters and the distribution of permanent productivity conditional on health type using the method of simulated moments. We first describe the inputs constructed in each step and the variation informing the parameters, then assess the model’s fit to lifecycle employment, healthy choices, health, and earnings.

5.1 First step: earnings, health, and initial conditions

The first step constructs the environment in which individuals make employment, saving, and healthy-choice decisions. We estimate the earnings and health processes directly from the panel, construct the initial distributions, and fix the remaining external parameters. These inputs are held constant when estimating the structural parameters in the second step.

The deterministic earnings component. We estimate the predictable component of log annual labor earnings among employed individuals aged 25–60 using survey weights. The regression contains a cubic in age, indicators for composite health with *GG* as the reference, and controls for urban residence, race, marital status, sex, education, and survey year. The age coefficients determine $g(24 + j)$ in equation (3), and the health coefficients determine λ_h . The demographic and year controls isolate these profiles in the data; they do not become additional state variables in the model. The model’s earnings level is determined by the type-specific means of permanent productivity estimated in the second step, so the regression intercept is not added separately to those means. Appendix B.1 reports the specification, the estimates, and the implied lifecycle profiles of earnings by health state.

The stochastic earnings component. Let r_{it} be individual i ’s observed log earnings in survey wave t minus the fitted value from the regression above, including its controls. This residual measures earnings variation left unexplained by the observed characteristics. Repeated observations reveal how much of that variation persists: we calculate the covariance between a person’s residual in wave t and their residual k annual waves earlier,

$$C(k) \equiv \text{Cov}(r_{it}, r_{i,t-k}), \quad k = 0, 1, \dots, 8. \quad (8)$$

The index k is the lag: $C(0)$ is the residual variance, $C(1)$ measures persistence over one annual wave, and larger values of k measure persistence over longer intervals. Each empirical covariance uses observations eligible at both endpoints, weighted by the survey weight of the current observation.

We interpret these covariances through the decomposition

$$r_{it} = a_i + z_{it} + e_{it}, \quad z_{it} = \rho z_{i,t-1} + \nu_{it}.$$

The component a_i is permanent, z_{it} is persistent but mean-reverting, and e_{it} is serially independent. Under mutual orthogonality of the components and stationarity of z_{it} , their implied covariance function is

$$C(k) = A + B\rho^k + E \mathbf{1}\{k = 0\}, \quad A = \text{Var}(a_i), \quad B = \text{Var}(z_{it}), \quad E = \text{Var}(e_{it}). \quad (9)$$

The permanent component contributes A at every lag. The persistent component contributes $B\rho^k$, which declines with the lag, while the independent component contributes only to $C(0)$. The decline across positive lags therefore informs ρ and B , the covariance remaining at longer lags informs A , and the additional variance at lag zero informs E . We estimate all four parameters jointly by minimizing equally weighted squared differences between empirical and implied covariances at lags zero through eight, imposing nonnegative variance components. Appendix B provides the identification argument and implementation details.

The estimates are $\rho = 0.87217$, $A = 0.07168$, $B = 0.08896$, and $E = 0.02439$. The implied innovation variance is $\sigma_\nu^2 = B(1 - \rho^2) = 0.02129$. Only ρ and σ_ν^2 enter the structural earnings process.⁹ We also exclude E from the simulation because this residual component does not distinguish economic risk from measurement error. Finally, the external decomposition uses stationarity to estimate the law of motion, whereas the model initializes $z_1 = 0$: initial earnings dispersion is captured by health and permanent productivity, and persistent shocks accumulate subsequently. These estimates describe earnings observed among workers; the external regression does not separately correct for employment selection.

Permanent health types and initial health. We construct the four permanent health types from residualized mental and physical health histories as described in Section 3.2.1. Survey-weighted type frequencies at age 25 determine the initial type distribution. Within each type, the weighted distribution of composite health at age 25 determines initial health. The type-specific distribution of permanent productivity is left to the second step, allowing initial health and earnings capacity to be associated.

Initial healthy choices. Because changing a healthy choice is costly, each individual enters the model with a pair of previous choices $F_1 = (f_0^B, f_0^C)$. We draw it from the weighted joint distribution of (f^B, f^C) observed at ages 25–29, separately for the lower and upper halves of the residual labor-earnings distribution used for the income moments. In the model, the half is assigned by permanent productivity θ , which is the residual component of log earnings at entry once age and health are removed. Among the lower half, 38.3% make the healthy-consumption choice at ages 25–29, compared with 51.7% in the upper half; the corresponding healthy-behavior shares are 50.7% and 55.2%. These initial differences are an input, not an outcome of the model; the switching costs estimated in the second step govern how long they persist.

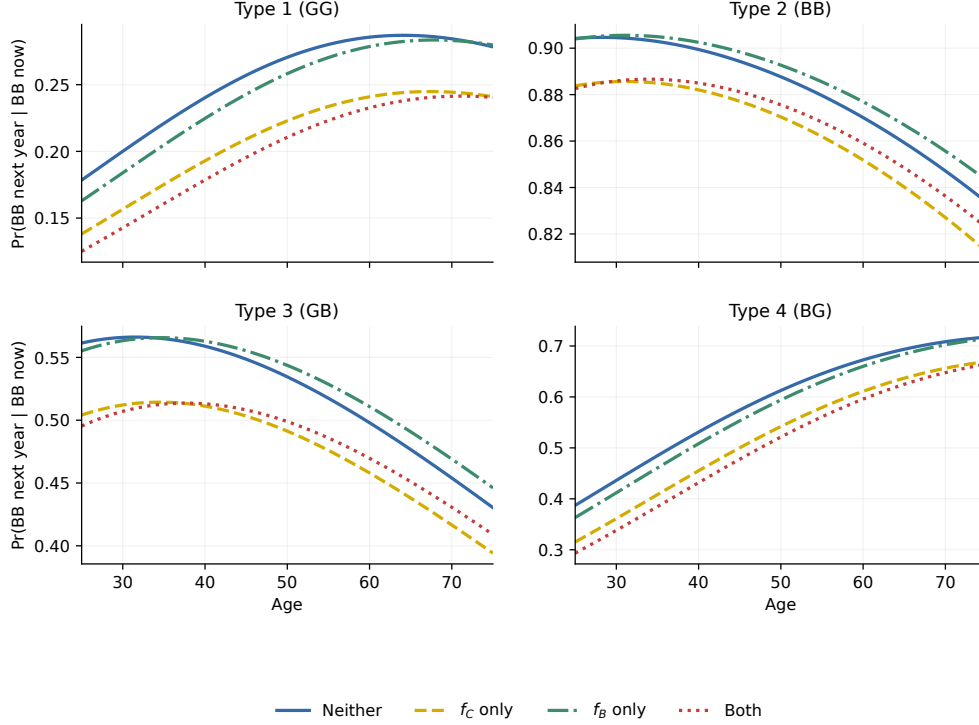
Health transitions. The survey-weighted multinomial logit predicts next-period composite health conditional on age, current composite health, both healthy choices, and permanent health type. We then adjust its predicted probabilities to match one-year state retention and age-specific health shares. The adjustment first changes origin-state retention and then the distribution of destination states, preserving valid probability rows. Appendix C describes this recalibration. The resulting transition law is held fixed during the second-step estimation.

Figure 9 illustrates how healthy choices and permanent types enter the estimated transition law. It plots one of its central objects, the predicted probability that an individual in BB remains in BB the following year. Each panel corresponds to a permanent health type, and the four lines compare the four combinations of healthy behavior and healthy consumption over age. The distance between lines measures how the probability of remaining in the worst health state varies with healthy choices, holding age and type fixed; the differences across panels measure how much permanent type shifts that persistence. At age 50, for example, the probability is 0.27 for a type-1 individual who makes neither choice and 0.21 for one who

⁹Permanent productivity is estimated separately, conditional on health type, so adding an independent draw with variance A would count permanent heterogeneity twice.

makes both, whereas for type 2 it is 0.89 and 0.88. The figure shows the multinomial-logit predictions before the recalibration described above.

Figure 9: Probability of remaining in BB



Notes: Predicted probability by permanent health type and healthy choices, before transition recalibration.

Survival. Survival probabilities $s_j(h)$ depend on age and current composite health, and we construct them in two steps. First, we estimate a logit model for the probability of dying before the next wave, recorded in the UKHLS as a death outcome at the following interview, on age and indicators for composite health, with GG as the reference. Deaths are recorded only for sample members who remain in contact with the survey, so the predicted mortality is too low relative to the population. Second, we correct its level with an age-specific shift in the log-odds of death, common to the four health states, chosen so that at each age the average mortality across health states, weighted by their shares in the UKHLS, equals the mortality rate in the life tables of the Office for National Statistics for the United Kingdom. The correction matches aggregate mortality by age while preserving the differences in mortality across health states estimated in the first step. Appendix C.2 reports the estimates and the correction. Because survival depends on current health, healthier choices can affect survival through subsequent health transitions. The survival schedule is not re-estimated with preferences and costs.

Assets and fixed preference parameters. Initial assets are £21,085, the mean net wealth at ages 20–24, used as an estimate of wealth upon entry at age 25. Individuals

cannot borrow. We fix the discount factor at $\beta = 0.96$, the interest rate at $r = 0.03$, and the consumption share at $\alpha = 0.60$. Utility curvature is set to $\gamma = 2$, a standard value for intertemporal substitution; with $\alpha = 0.60$ it implies a curvature of $1 + \alpha(\gamma - 1) = 1.6$ in consumption. The flow value of being alive is $b = 0.011$, calibrated so that a year of life is worth about four years of consumption, as in Hall and Jones (2004). These preference restrictions are maintained calibrations, rather than parameters separately identified by the moments below.

5.2 Second step: the method of simulated moments

Taking the first-step inputs as given, we estimate 41 parameters governing employment and healthy-choice costs, the costs of changing healthy choices, the discrete-choice shock scale, and permanent productivity conditional on health type. The utility flow from being alive, $b = 0.011$, and the utility curvature, $\gamma = 2$, are held fixed rather than estimated. For each candidate parameter vector, we solve the individual's dynamic problem, simulate lifecycle histories, and calculate the same moments as in the data. Employment selection is therefore determined within the model when constructing simulated earnings moments.

Parameters and identifying variation. Table 2 links each parameter or indexed parameter family to its economic role and the variation that informs it. It also records external inputs and fixed calibrations. Identification is joint: changes in one estimated parameter can affect several moment blocks, and no single moment isolates a parameter independently of the rest of the model. In particular, the time cost and direct utility cost of healthy behavior can substitute for one another, and the monetary price of healthy consumption interacts with its switching cost: persistence makes a given price matter more for the choice, because adopting healthy consumption commits resources over several years. The flow value of being alive, by contrast, is held fixed in the reported estimation; it affects choices through the continuation value of survival but is not identified by the simulated moments.

Table 2: Parameters, economic roles, and identifying information

Parameter	Economic role	Moments or external calibration
<i>Second-step parameters (41 coefficients)</i>		
$\psi_0^n, \psi_1^n, \psi_2^n, \psi_3^n$	Level, slope, curvature, and cubic term of the employment cost in GG	Employment shares across age groups in GG , including the late-career decline.
$\Delta_h^n,$ $h \in \{BB, BG, GB\}$	Health-specific employment-cost intercepts	Employment-level differences across health states.
$\psi_{n,m}^h, m = 1, 2, 3;$ $h \in \{BB, BG, GB\}$	Health interactions with linear, quadratic, and cubic age terms	Differences in employment age profiles across health states.
$\psi_0^B, \psi_1^B, \psi_2^B$	Level, slope, and curvature of the healthy-behavior utility cost	Healthy-behavior shares across age groups.

Table 2 (continued)

Parameter	Economic role	Moments or external calibration
Δ_h^B , $h \in \{BB, BG, GB\}$ $\psi_0^C, \psi_1^C, \psi_2^C$	Health-specific behavior-cost intercepts Level, slope, and curvature of the healthy-consumption utility cost	Differences in healthy-behavior shares across health states. Healthy-consumption shares across age groups.
Δ_h^C , $h \in \{BB, BG, GB\}$ ϕ_B	Health-specific consumption-cost intercepts Time cost of healthy behavior	Differences in healthy-consumption shares across health states. Behavior shares by age and health, and above/below the residual-income median, jointly with utility costs.
pf	Annual monetary cost of healthy consumption, paid whether or not the individual works	Consumption shares, especially their difference across the residual-income median conditional on the previous consumption choice.
κ_B, κ_C	Utility costs of changing healthy behavior and healthy consumption relative to the previous period	Shares making each choice among those who did and did not make it two years earlier.
τ	Scale of discrete-choice taste shocks	Employment and healthy-choice shares jointly, through their responsiveness to utility differences.
$\mu_{\theta,k}$, $k = 1, 2, 3, 4$	Type-specific means of permanent log productivity	Mean log earnings at ages 25–29 within each health type, accounting for employment selection.
$\sigma_{\theta,k}$, $k = 1, 2, 3, 4$	Type-specific standard deviations of permanent log productivity	Within-type log-earnings variances at ages 25–29 and pooled variances at each age 25–29.
<i>First-step estimates and externally constructed inputs</i>		
$g(\cdot)$	Cubic age profile of offered log earnings	Survey-weighted log-earnings regression among workers aged 25–60.
$\lambda_h, \lambda_{GG} = 0$	Health-specific earnings differences	Composite-health coefficients in the same regression.
ρ, σ_ν^2	Persistence and innovation variance	Residual covariances $C(0), \dots, C(8)$; $\sigma_\nu^2 = B(1 - \rho^2)$.
A, B, E	Permanent, stationary persistent, and independent residual variances	Joint covariance decomposition; A and E are not added to structural earnings.
$\Pr(\eta = k)$, $\Pr(H_1 = h \mid \eta = k)$	Initial health-type and within-type health distributions	Weighted type and health frequencies at age 25.

Table 2 (continued)

Parameter	Economic role	Moments or external calibration
$\pi_j(\cdot)$	Conditional health-transition probabilities	Multinomial logit; recalibration to one-year retention and age-specific health shares.
$s_j(h)$	Survival probabilities	Logit of mortality on age and composite health, with an age-specific level correction to ONS life tables.
<i>Fixed calibrations and initial conditions</i>		
$\beta = 0.96$	Discount factor	Fixed calibration.
$r = 0.03$	Interest rate	Fixed calibration.
$\alpha = 0.60$	Consumption share in utility	Fixed calibration.
$\gamma = 2$	Utility curvature	Fixed calibration; curvature in consumption $1 + \alpha(\gamma - 1) = 1.6$.
$b = 0.011$	Utility flow from being alive	Calibrated so that a year of life is worth about four years of consumption (Hall and Jones, 2004); it receives no structural standard error.
$a_1 = \text{\pounds}21,085$	Initial net assets	Mean net wealth at ages 20–24, used to approximate wealth at age 25.
$\underline{a} = 0$	Borrowing limit	No borrowing.
$F_1 = (f_0^B, f_0^C)$	Previous healthy choices at entry	Weighted joint distribution at ages 25–29 by half of the residual labor-earnings distribution, assigned in the model by half of θ .
$z_1 = 0$	Initial persistent earnings component	Normalization; initial dispersion is assigned to health and permanent productivity.

Notes: Indexed rows cover every coefficient in the indicated family. All age polynomials use normalized model age. Health-cost differences are relative to GG . The moment mappings describe joint identifying variation, not independent one-to-one identification.

Empirical moments. The estimation targets $M = 145$ moments, all computed from the UKHLS with longitudinal survey weights and computed identically from the simulated panel. They form five groups; Table D.1 in Appendix D lists their indices.

1. *Employment and healthy-choice profiles (120 moments).* For each of the four composite health states and each of the ten five-year age groups from 25–29 to 70–74, the share of person-years employed (40 moments), the share making the healthy-behavior choice (40), and the share making the healthy-consumption choice (40).
2. *Income gradient (4 moments).* Among person-years aged 25–60 with positive labor earnings and both healthy choices observed, we regress log labor earnings on a quadratic

in age and indicators for composite health and split the sample at the weighted median of the residual. The moments are the shares making the healthy-behavior choice and the healthy-consumption choice in the upper and in the lower half.

3. *Early-adult earnings (13 moments)*. Among employed individuals aged 25–29, the mean and the variance of log labor earnings within each permanent health type (8 moments), and the variance of log labor earnings at each single age from 25 to 29, pooling types (5 moments).
4. *Persistence (4 moments)*. Among person-years aged 27–60 whose previous observation of the behavioral module lies one to three years earlier, the share making each healthy choice among those who made it at the previous observation and among those who did not. In the model, the previous observation is the choice made two years earlier.
5. *Income gradient conditional on the previous choice (4 moments)*. In the same sample, the share making the healthy-consumption choice in the upper and in the lower half of the residual-earnings distribution defined in group 2, separately for those who did and did not make the healthy-consumption choice at the previous observation.

The first group disciplines the age and health profiles of the employment and healthy-choice costs. The persistence moments discipline the switching costs, and the two income-gradient groups discipline the monetary cost of healthy consumption: conditioning on the previous choice separates the part of the gradient carried by past choices from the part related to current earnings. The early-adult earnings moments discipline the level and dispersion of permanent productivity within each health type, while the first-step shock process governs how earnings dispersion grows with age.

Estimation criterion. Let m^D denote the empirical moments and $m^S(\vartheta)$ their simulated counterparts, with $\vartheta \in \mathbb{R}^{41}$. We estimate

$$\hat{\vartheta} = \arg \min_{\vartheta} Q(\vartheta), \quad Q(\vartheta) = [m^S(\vartheta) - m^D]' \hat{\Omega}^{-1} [m^S(\vartheta) - m^D]. \quad (10)$$

Here $\hat{\Omega}$ is the sampling variance–covariance matrix of the empirical moments. We estimate it by an individual-cluster bootstrap with 2,000 replications, resampling the 26,853 individuals with their complete observed histories and recomputing all moments. This preserves within-person dependence and accounts for covariance among moments based on overlapping respondents. The matrix is computed before structural estimation and held fixed throughout. The criterion contains no additional penalty or regularization term.

We minimize this distance using a finite-difference Jacobian, simulating 50,000 lifecycles per evaluation. Because the criterion admits distinct local minima in the price of healthy consumption, we started the optimization from several values of p_f and κ_C and report the solution with the lowest criterion. Appendix D describes the numerical solution, optimization, and convergence diagnostics. Table D.2 reports parameter estimates and sandwich standard errors that combine the bootstrap covariance of data moments with simulation covariance estimated across independent replications. These standard errors condition on the first-step inputs. Because the weighting matrix includes data sampling uncertainty but not simulation

variance, it is inverse-sampling-variance weighting and need not be asymptotically efficient. We use $Q(\hat{\vartheta})$ as an estimation objective rather than as a formal specification test.

5.3 Model fit

We assess how closely the estimated model reproduces the observed evolution of health, employment, healthy choices, and earnings. We begin with health composition and persistence, then examine each labor-market and choice profile in turn. Finally, we evaluate the evolution of earnings dispersion beyond the early-adult moments used in estimation.

5.3.1 Health, employment, earnings, and healthy choices

Figure 10 evaluates both the composition and persistence of health. Panel (a) shows that the model reproduces the main lifecycle shifts: isolated mental-health deficits become less prevalent, physical-health deficits become more prevalent, and the share in joint good health declines. For example, between ages 25 and 70 the simulated *BG* share falls from 35.0% to 8.2%, compared with 35.0% to 7.9% in the data; the simulated *GB* share rises from 10.0% to 37.6%, compared with 10.0% to 39.0%. Panel (b) shows that the model also tracks the decline in same-state retention as the lag increases, although its decline is more pronounced. At lag one, retention is 0.676 in the model and 0.589 in the data; at lag five it is 0.463 and 0.522, respectively. Retention measures the probability of occupying the same composite state at two observations separated by the indicated lag.

Figure 10: Health composition and persistence: model versus data

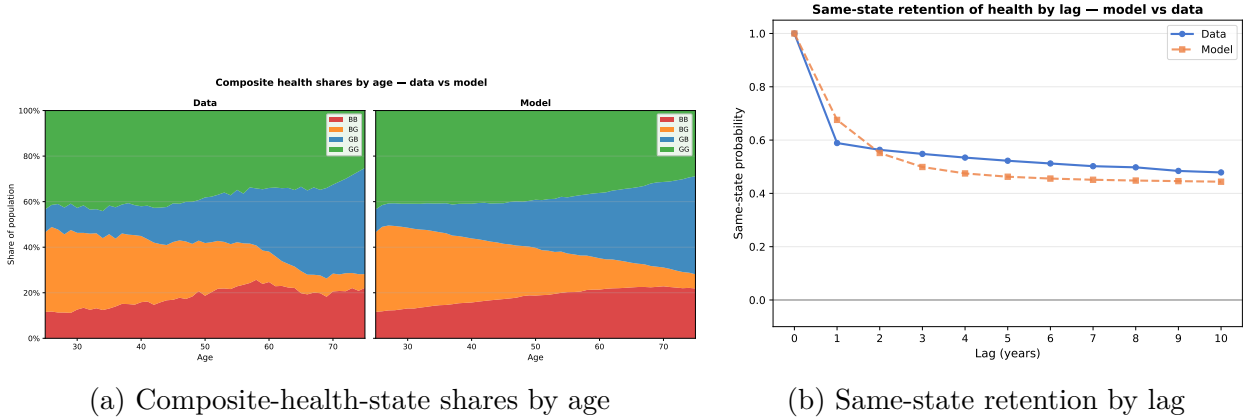
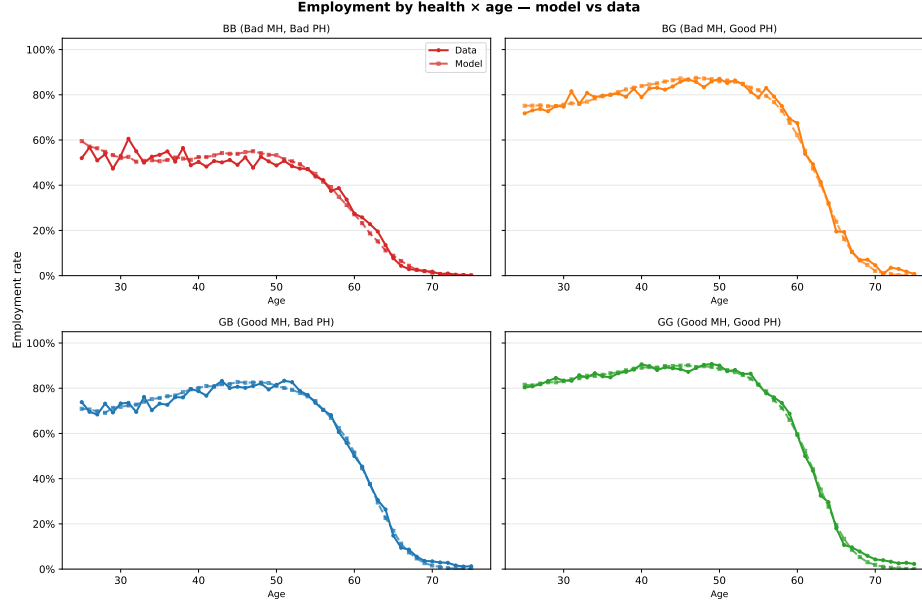


Figure 11 compares employment profiles across health states. The model captures both the lower participation associated with joint health deficits and the sharp decline in employment at later ages. Between ages 60 and 70, employment in *GG* falls from 59.7% to 1.9% in the model, compared with 59.3% to 4.3% in the data; in *BB*, the corresponding declines are from 27.1% to 1.3% and from 27.4% to 1.8%. The health-specific age profiles of employment costs allow the timing and level of this decline to differ across states.

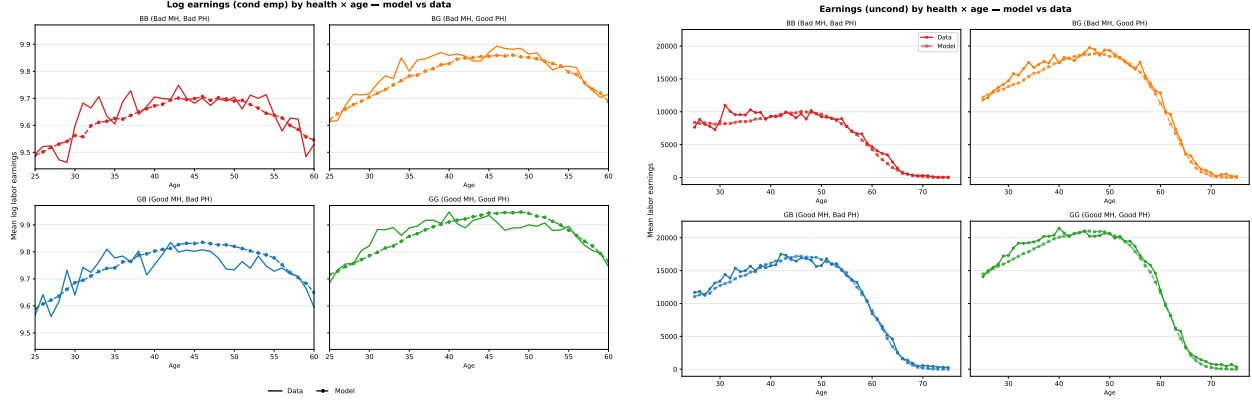
Figure 11: Employment by composite health state: model versus data



Notes: Employment rate by age.

Figure 12 compares earnings by health state. Panel (a) shows mean log earnings among workers. The model reproduces the broad hump-shaped age profiles and the lower earnings associated with joint health deficits. At age 50, mean log earnings in *GG* are 9.94 in the model and 9.90 in the data, while in *BB* they are 9.69 and 9.70. The fit is less close for isolated physical deficits, for which the model predicts 9.82 compared with 9.73. These profiles reflect both offered earnings and endogenous selection into employment. Panel (b) adds the participation margin by including zero earnings among nonworkers. The model captures the rise and subsequent decline in unconditional earnings and the large gap between *BB* and *GG*: at age 50, it predicts £9,628 in *BB* and £20,642 in *GG*, compared with £9,243 and £20,598 in the data.

Figure 12: Labor earnings by composite health state: model versus data



(a) Log earnings, conditional on employment

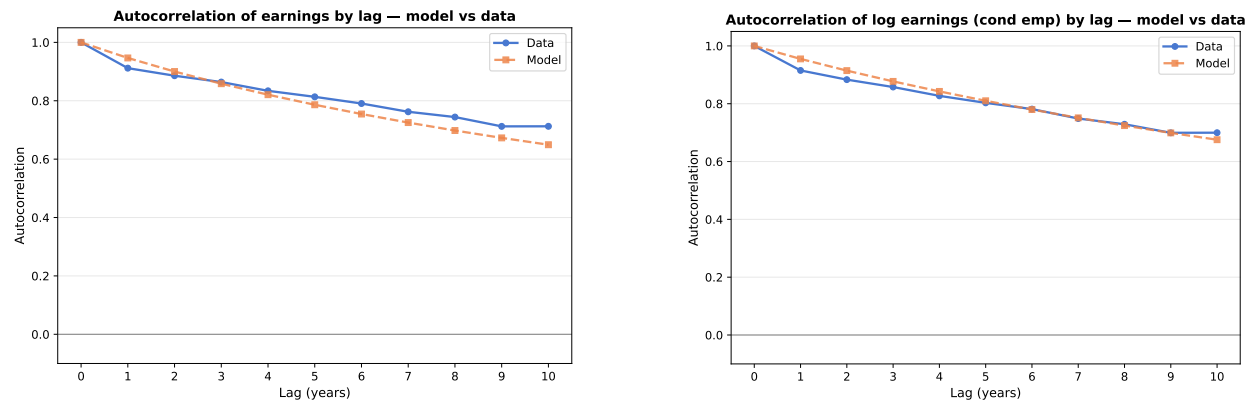
(b) Unconditional earnings

Notes: Panel (a) reports mean log labor earnings among the employed, ages 25–60; solid lines show data and dashed lines with dots show the model. Panel (b) reports mean labor earnings by age, including zero earnings for individuals not employed.

The late-career decline in unconditional earnings reflects both falling employment and lower conditional earnings.

Figure 13 assesses earnings persistence using autocorrelations in levels and in logs conditional on employment. The model tracks the decline in earnings correlation as the lag increases: at lag one it is 0.947 in the model and 0.912 in the data, while at lag five the values are 0.786 and 0.813. These autocorrelations are not targeted in estimation. They reflect the persistence and innovation variance of the stochastic earnings component, estimated in the first step from the residual covariances; the dispersion of permanent productivity within health types, disciplined by the early-adult earnings means and variances; and the persistence of health and employment generated by the model.

Figure 13: Earnings autocorrelations: model versus data



(a) Earnings

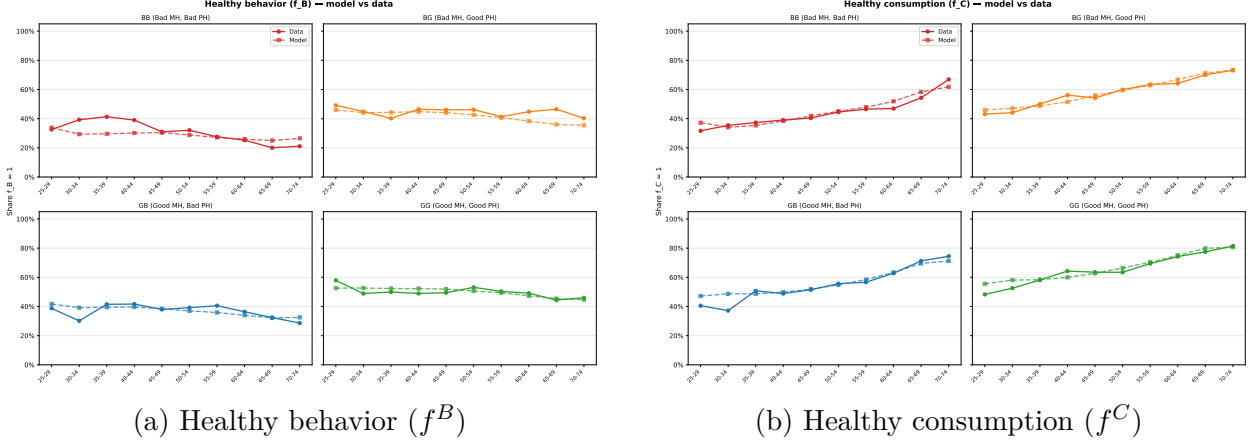
(b) Log earnings (cond. emp.)

Notes: Autocorrelation by lag of labor earnings (panel a) and of log labor earnings conditional on employment (panel b).

Figure 14 compares healthy-choice shares across five-year age groups and health states.

The model reproduces the broad decline in healthy behavior and the increase in healthy consumption with age, together with lower participation in both choices among individuals in *BB* than in *GG*. For *GG*, simulated healthy consumption rises from 55.6% at ages 25–29 to 80.7% at ages 70–74, compared with 48.3% to 81.5% in the data. Healthy behavior falls from 52.6% to 44.7% over the same ages, compared with 57.9% to 46.0%. The estimated costs capture these distinct profiles, although some age–health cells remain less well matched.

Figure 14: Healthy choices by composite health state: model versus data



Notes: Targeted moments in five-year age bins.

Table 3 reports the four income-gradient targets. The model reproduces the healthy-consumption gradient: 9.32 percentage points compared with 8.09 in the data, with consumption shares of 59.84% above and 50.52% below the median, compared with 59.11% and 51.01%. It slightly overstates the gradient in healthy behavior, 2.79 points compared with 0.45. The consumption gradient is generated by the combination of the monetary cost of healthy consumption, which weighs more on lower resources, and the persistence of choices, which carries initial differences forward and amplifies the effect of a given price. These discrepancies matter for interpreting both income-targeted interventions and the price experiment.

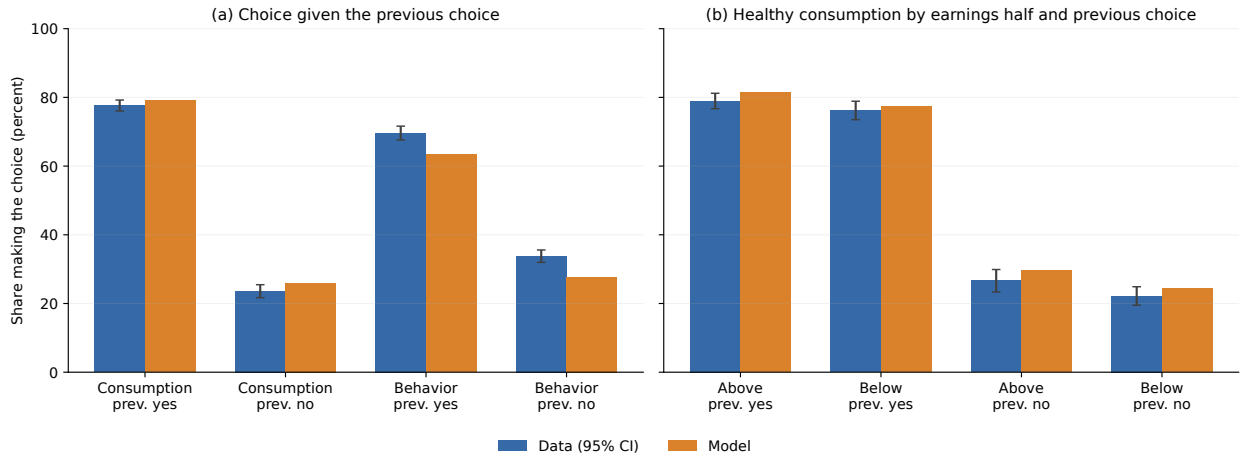
Table 3: Income gradient in healthy choices: model versus data

	Model	Data	Data s.e.
<i>Healthy behavior</i>			
Above the median	46.38	47.38	0.92
Below the median	43.59	46.93	0.90
Difference	2.79	0.45	1.25
<i>Healthy consumption</i>			
Above the median	59.84	59.11	0.96
Below the median	50.52	51.01	0.92
Difference	9.32	8.09	1.26

Notes: Percent of person-years making each healthy choice above and below the weighted median of residual log labor earnings (age, age squared, and composite health removed), ages 25–60; differences in percentage points. Standard errors come from the individual-cluster bootstrap used to construct the weighting matrix.

Figure 15 reports the persistence moments. The model reproduces the persistence of healthy consumption closely: 79.1% of those who made the choice two years earlier make it again, compared with 77.6% in the data, against 26.0% and 23.6% among those who did not. It understates the persistence of healthy behavior, mainly because too few individuals who did not make the choice take it up. Conditional on the previous consumption choice, the model generates income gradients of 4.1 and 5.3 percentage points, compared with 2.8 and 4.4 in the data.

Figure 15: Persistence moments: model versus data

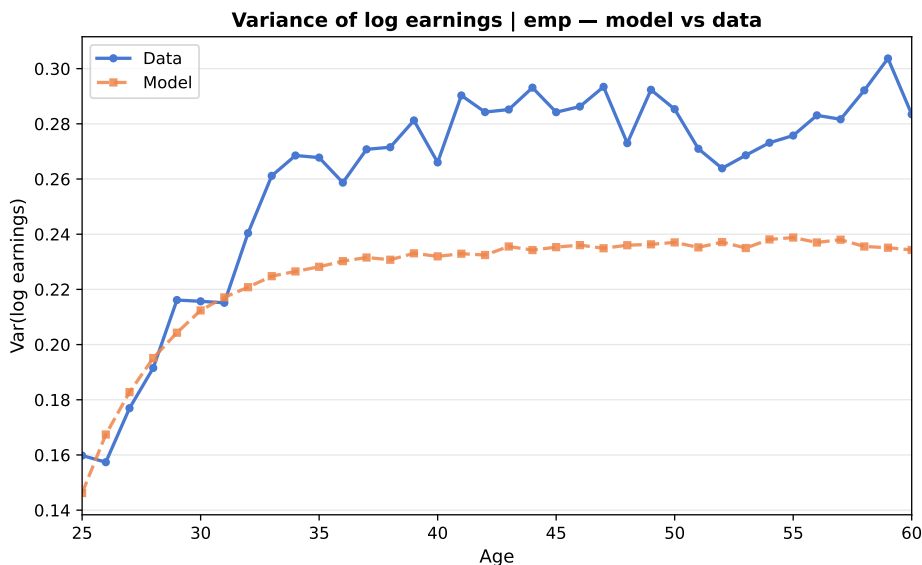


Notes: Ages 27–60. Panel (a) reports the share making each choice among those who did and did not make it at the previous observation of the behavioral module, two years earlier. Panel (b) splits the healthy-consumption share at the median of residual labor earnings, by the previous healthy-consumption choice. Whiskers are 95% confidence intervals for the data, from the individual-cluster bootstrap.

5.3.2 Earnings dispersion beyond the initial-age targets

The age profile of earnings dispersion is a useful validation exercise because the model does not target a separate variance at every age. Nevertheless, it is not entirely untargeted: variances at ages 25–29 enter the structural criterion. Figures 16 and 17 focus on ages 25–60 to avoid the terminal retirement boundary and describe both total dispersion and its distribution across tails. Figure 16 shows that the model reproduces early working-life dispersion relatively closely but understates its increase between ages 30 and 35: over those ages, the observed variance rises by 0.052 and the simulated variance by 0.016. After age 35, both profiles are essentially flat. Between ages 35 and 60, the observed variance rises by 0.016 and the simulated variance by 0.006, so the model reproduces the stagnation of dispersion after age 35, at a lower level. At ages 30, 40, 50, and 60, model log-earnings variances are 0.212, 0.232, 0.237, and 0.234, compared with 0.216, 0.266, 0.285, and 0.283 in the data. The gap that opens between ages 30 and 35 points to limits in how the estimated earnings process and employment selection jointly reproduce the early growth of dispersion beyond the initial-age targets.

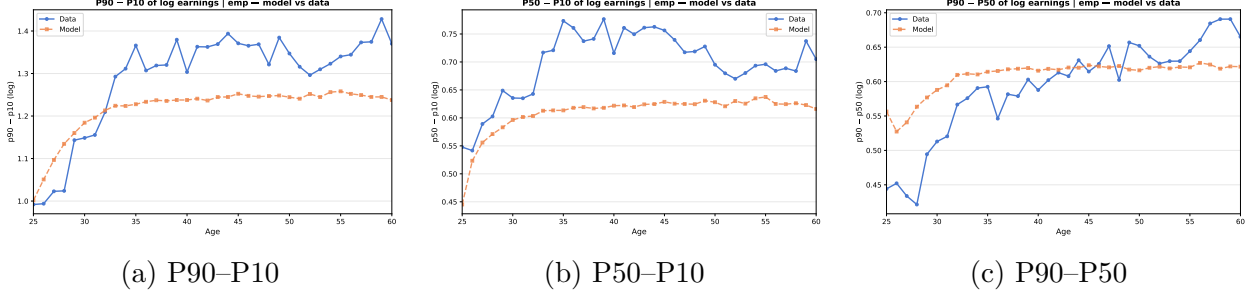
Figure 16: Variance of log labor earnings: model versus data



Notes: Conditional on employment, ages 25–60. Variances at ages 25–29 enter estimation; the profile at ages 30–60 is not directly targeted.

Figure 17 examines the corresponding percentile gaps, separating the upper and lower halves of the earnings distribution. The simulated P90–P10 log difference is 1.184 at age 30, compared with 1.149 in the data; by age 60 it is 1.238, compared with 1.370. The percentile comparison therefore also indicates that the model generates too little widening of earnings differences over working life.

Figure 17: Percentile log differences in labor earnings: model versus data



Notes: Log labor earnings conditional on employment, ages 25–60.

6 Results: Healthy choices and lifetime earnings inequality

We ran a counterfactual analysis to study the implications of healthy choices for health and earnings dispersion. The experiments share individual random draws with their paired baseline simulations to reduce simulation noise and maintain the estimated conditional health-transition mapping when choices are imposed. They therefore evaluate the mandates under the assumption that the transition probabilities remain invariant to the intervention. The principal inequality outcome is the variance of log cumulative labor earnings from ages 25 through 55, among individuals alive at age 55 in both simulations and with positive cumulative earnings in the relevant scenario. We use “lifetime earnings” as shorthand for this working-age accumulation window, rather than the full age-25–75 model horizon. Appendix E defines the samples and aggregation precisely.

We consider three groups of experiments: health-based mandates, income-based mandates, and a price-based experiment. In every experiment, we solve the Bellman problem again under the intervention, so individuals re-optimize employment, consumption, and saving, anticipating the intervention’s implications for their future choices and resources. The mandates impose one or both healthy choices on the individuals covered by a rule by restricting their admissible choice set. They require the choice but do not provide it: an individual required to make the healthy-consumption choice pays its monetary cost p_f , and an individual required to make the healthy-behavior choice bears its time cost. Because the model has no borrowing and no transfers, meeting a consumption mandate requires resources.¹⁰

Health-based mandates. These experiments condition the mandate on current health: the restriction applies in every year in which an individual occupies a health state named by the rule, and it lapses when the individual leaves that state. Three experiments target *BB*, the state with the largest labor-market penalties. The first requires both healthy

¹⁰In the experiments that require healthy consumption, 85 of the 50,000 simulated individuals (0.17%) have no feasible path under the mandate: in some future state reached with positive probability their earnings cannot cover the cost of healthy consumption and consumption. We exclude them from both the baseline and the counterfactual simulation, so the comparison remains between the same individuals. No individual is excluded in the healthy-behavior mandate or in the price experiment.

choices whenever an individual is in BB ; the second requires only healthy behavior and the third only healthy consumption, which separates the contribution of each choice. The fourth experiment broadens the first to every state other than GG , so that any health deficit triggers the mandate. Finally, the universal mandate requires both choices in every health state and at every age. It is not targeted, and it provides an upper benchmark for the health gains that the estimated transitions allow.

Income-based mandates. These experiments condition the same requirement—both healthy choices—on earnings capacity. We first calculate P10, P25, and P50 monetary cut-offs from the pre-intervention distribution of positive annual labor earnings in the baseline simulation, separately at each age. These age-specific thresholds are fixed before solving the counterfactual and are not recalculated from its resulting income distribution. Eligibility is determined by offered annual earnings—the labor earnings the individual would receive if employed in the current state—rather than by realized earnings: both healthy choices are mandatory in every year in which offered earnings are at or below the cutoff, whether or not the individual works. Eligibility therefore cannot be changed through labor supply, but it changes over the life cycle as offered earnings move with age, health, and earnings shocks. A rule based on realized earnings would make every nonworker eligible and would let individuals with high offered earnings escape the mandate by working; Appendix E shows that this escape response would dominate the distributional effects of the rule.

Price-based experiment. The last experiment imposes no choice. It permanently reduces the monetary price of healthy consumption, p_f , to one tenth of its estimated value for everyone, and leaves individuals free to decide whether to adopt healthy consumption at the new price.

6.1 Healthy choices and lifecycle health

In the baseline, the share in the GG health state is 0.379, measured across alive person-years over ages 25–75. The remaining 0.621 are spent with a deficit in at least one health dimension. We examine how this composition changes when healthier choices are imposed according to health or income, and when healthy consumption becomes less costly. Figure 18 summarizes the resulting percentage changes in the share outside GG .

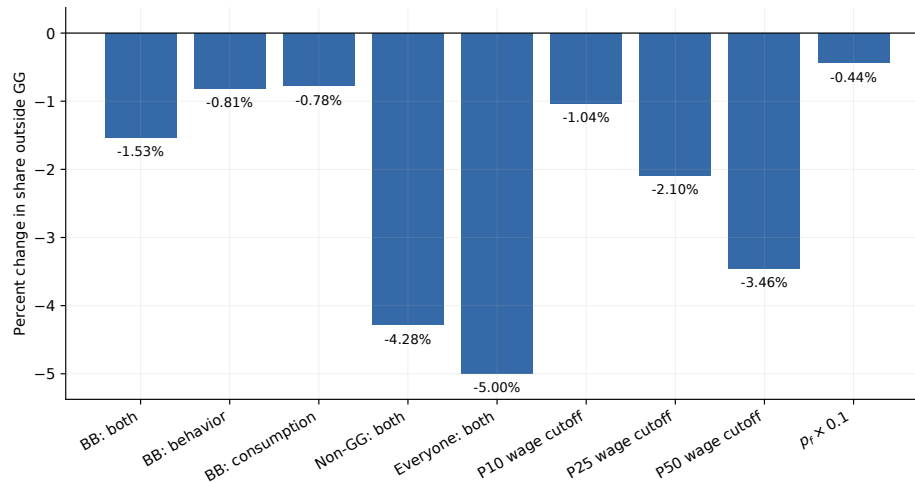
Among the health-based mandates, imposing both choices in BB reduces the share outside GG by 1.53%. Behavior alone reduces it by 0.81% and consumption alone by 0.78%. These effects need not add to the joint intervention because choices change future health and hence subsequent exposure to the rule. Broadening the mandate to everyone outside GG produces a larger reduction of 4.28%. Imposing both choices universally delivers the largest aggregate health improvement among the reported experiments: a 5.00% reduction, from 0.621 to 0.590.

Among the income-based mandates, expanding the threshold from P10 to P25 and then P50 increases the reduction in the share outside GG from 1.04% to 2.10% and 3.46%, respectively. These improvements remain smaller than under universal imposition, in line with the coverage of the rules: in the baseline, 9.96%, 25.18%, and 50.14% of alive person-years have

offered earnings at or below the respective cutoffs. Because eligibility does not depend on employment, these shares barely change in the counterfactual (9.83%, 24.97%, and 49.87%).

Finally, the price-based experiment encourages healthy consumption without imposing it. The resulting decline in the share outside *GG* is 0.44%, smaller than under the mandates. Because healthy consumption is persistent, a lower price raises adoption gradually rather than at once, and many of the individuals who adopt it were close to indifferent in the baseline.

Figure 18: Counterfactual effects on time outside good health



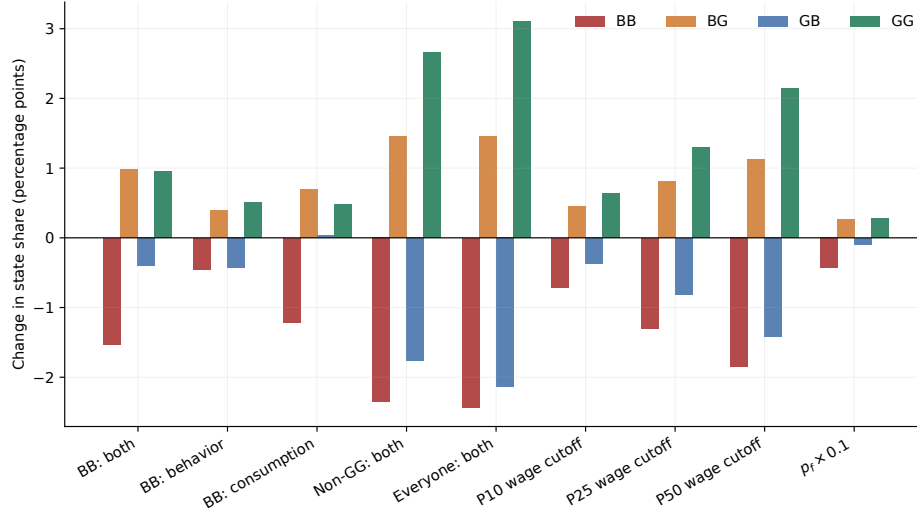
Notes: Percentage change relative to baseline in the share of alive person-years outside good mental and physical health (*GG*), ages 25–75. Baseline is 0.621. Negative values indicate a lower share outside *GG*.

A lower share outside *GG* does not reveal which health deficits the experiments remove. Figure 19 decomposes the change across the four states, in percentage points; in the baseline, 18.1% of alive person-years are spent in *BB*, 21.8% in *BG*, 22.2% in *GB*, and 37.9% in *GG*. Two patterns stand out. First, the experiments remove mainly physical-health deficits. Under universal imposition, the combined share of the two states with poor physical health, *BB* and *GB*, falls by 4.57 percentage points, whereas the combined share of the two states with poor mental health, *BB* and *BG*, falls by only 0.97 points: the *BB* share falls by 2.43 points, but the *BG* share rises by 1.46 points. Many individuals who leave *BB* therefore recover their physical health while their mental health remains poor. This pattern mirrors the evidence in Section 4: both healthy choices are associated with subsequent physical health, while only healthy consumption is associated, more weakly, with subsequent mental health. The mandates targeted at *BB* show the same composition. When both choices are imposed in *BB*, the *BB* share falls by 1.53 points, 8.5% of its baseline level, and the freed person-years go almost equally to *BG* (0.99 points) and *GG* (0.95 points).

Second, these movements out of *BB* matter for earnings even when they end in *BG*. The labor-market penalties documented in Section 4 are concentrated in joint deficits: relative to *GG*, employment is 31.3 percentage points lower in *BB* but only 3.8 points lower in *BG*, and conditional annual earnings are 15.0 log points lower in *BB* but only 3.4 points lower in *BG*. Leaving *BB* for *BG* therefore removes most of the labor-market penalty of poor

health without any improvement in mental health. Because time in *BB* is concentrated among individuals with low earnings, these transitions also reach the bottom of the earnings distribution. This is why the *BB*-targeted mandates, which change aggregate health little, have the large distributional effects reported below. Because the denominator counts alive person-years separately in each simulation, the state shares can also reflect changes in survival.

Figure 19: Counterfactual effects on composite health-state shares



Notes: Change in the share of alive person-years spent in each composite health state, in percentage points.

6.2 Healthy choices and lifetime earnings inequality

Figure 20 reports the corresponding effects on the variance of log cumulative earnings; the paired baseline values round to 0.187 across the mandates. Imposing both choices in *BB* reduces the variance by 9.12%. Healthy consumption accounts for most of this reduction: imposing consumption alone lowers the variance by 7.90%, while behavior alone lowers it by 1.56%. Extending both choices to everyone outside *GG* lowers the variance by 13.06%, and universal imposition by 13.36%. Table 4 collects the main outcomes of every experiment.

Table 4: Main counterfactual results

Experiment	Outside GG (%)	Mean Y (%)	Var(log Y) (%)
BB : both choices	−1.53	2.01	−9.12
BB : behavior	−0.81	0.62	−1.56
BB : consumption	−0.78	1.46	−7.90
Non- GG : both choices	−4.28	3.94	−13.06
Everyone: both choices	−5.00	4.80	−13.36
Offered earnings $\leq$ P10	−1.04	0.86	−10.65
Offered earnings $\leq$ P25	−2.10	1.78	−14.74
Offered earnings $\leq$ P50	−3.46	2.86	−16.06
$p_f \times 0.1$	−0.44	−1.14	2.02

Notes: Changes relative to each experiment’s paired baseline. “Outside GG ” is the percent change in the share of alive person-years at ages 25–75 spent outside good mental and physical health (baseline 0.621). Y is cumulative labor earnings over ages 25–55 among individuals alive at 55 in both simulations (baseline mean £494,865; baseline variance of logs 0.187, or 0.190 for the behavior and price experiments).

Cumulative earnings rise most at the bottom of the distribution. Under universal imposition, mean cumulative earnings rise by 4.80%; among common survivors, they rise by 15.0% in the bottom decile of baseline cumulative earnings, 7.0% between P10 and P50, 3.7% between P50 and P90, and 2.0% in the top decile. The healthy-behavior mandate, whose only cost is time, improves health by as much as the consumption mandate in BB but compresses cumulative earnings by a fifth as much. Appendix Table E.2 reports the decomposition.

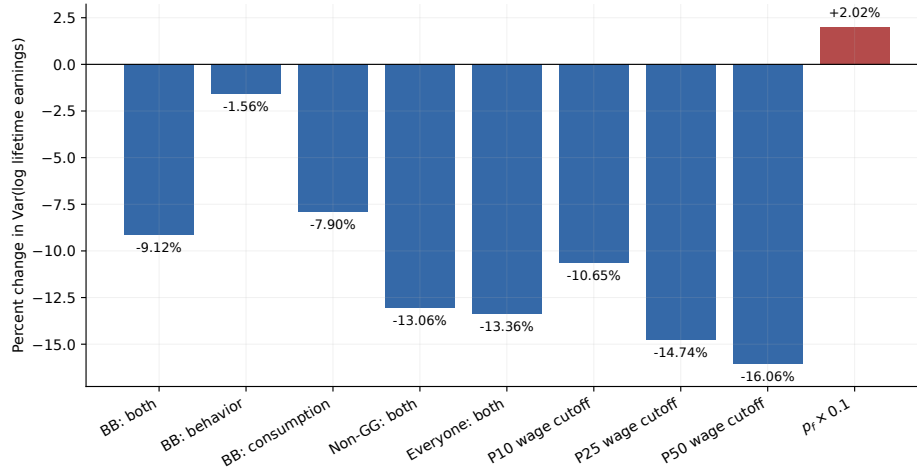
Targeting by offered earnings compresses cumulative earnings further. The P10, P25, and P50 rules reduce the variance by 10.65%, 14.74%, and 16.06%, respectively, more than universal imposition, although their health effects are between a fifth and seven tenths as large. Under the P50 rule, mean cumulative earnings rise by 14.6% in the bottom decile of baseline cumulative earnings, 5.5% between P10 and P50, 1.3% between P50 and P90, and 0.1% in the top decile.

The comparison between income-based and health-based mandates in Figure 20 speaks to how interventions should be targeted when the objective is to reduce earnings inequality. The two forms of targeting differ in how directly they reach the bottom of the earnings distribution. Health-based rules reach low earners only indirectly: time in BB is concentrated at the bottom of the distribution, but the rule also covers individuals in BB with high earnings capacity and misses low earners in better health. Income-based rules reach low earners directly. Measured relative to the share of person-years they cover, income-based rules compress cumulative earnings more. The P10 rule covers about 10% of alive person-years and reduces the variance by 10.65%, more than the BB mandate, which covers 18% of person-years and reduces it by 9.12%. The P10 rule also improves aggregate health by more per person-year covered, because low earners are disproportionately in poor health. Moreover, the compression achieved by health-based rules levels off as coverage widens: extending the mandate from every state other than GG to everyone raises the reduction only from 13.06% to 13.36%. Income-based rules, by contrast, keep compressing as the threshold rises, up to 16.06% under the P50 rule. If the objective is a more equal distribution of earnings,

targeting earnings capacity directly is more effective than targeting health; targeting health is preferable only insofar as the policy objective gives independent weight to health itself. These conclusions concern the targeting rule, not its implementation: offered earnings are not observed for nonworkers, so a practical rule would have to rely on observable proxies for earnings capacity, such as past earnings or education.

Reducing the monetary price of healthy consumption by 90% improves aggregate health modestly, lowering the share outside *GG* by 0.44%, but raises the variance of log cumulative earnings by 2.02%. The lower price makes healthy consumption more affordable without directly changing offered earnings at a given health and productivity state. Individuals respond through healthy consumption, saving, and employment: the share of alive person-years spent working falls from 58.94% to 57.92%, and mean cumulative earnings fall by 1.14%. The employment response is larger toward the lower end of the baseline cumulative-earnings distribution. Among common survivors, the work share over ages 25–55 falls by 1.27 and 1.32 percentage points in the bottom decile and between P10 and P50, compared with 0.92 points between P50 and P90 and 0.48 points in the top decile. The price reduction weakens the need to work to finance healthy consumption, most strongly where earnings are lowest, so the labor-supply response increases dispersion despite the modest health improvement. Appendix Table E.3 reports the decomposition.

Figure 20: Counterfactual effects on lifetime-earnings inequality



Notes: Percentage change relative to baseline in the variance of log cumulative earnings over ages 25–55, among individuals alive at age 55 in both simulations with positive cumulative earnings in each scenario. Income-contingent rules use offered earnings to determine eligibility.

6.3 Other sources of earnings inequality

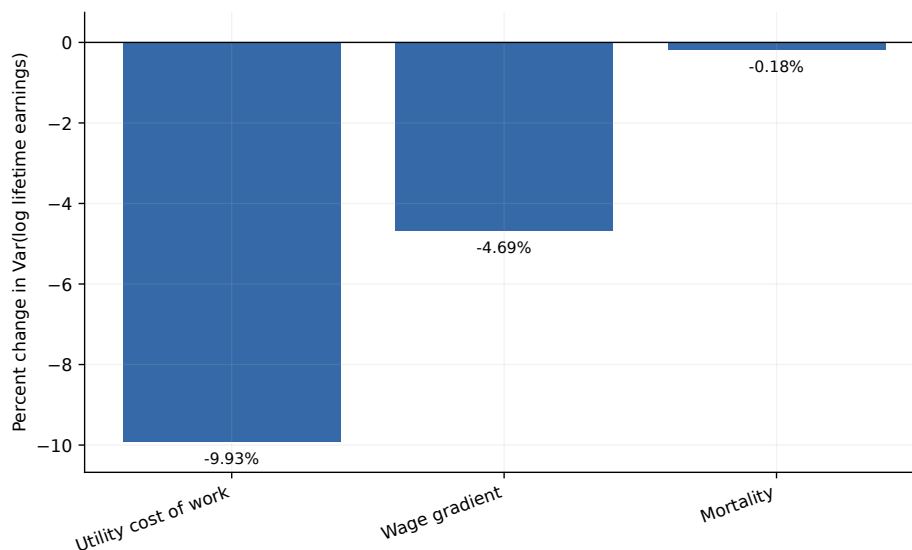
The imposed-choice experiments change which health states individuals reach. A separate question is how health reaches earnings once a state is realized. Health enters the model in three places: the offered-earnings gradient λ_H , the utility cost of working $\psi_j^n(H)$, and survival $s(j, H)$. Figure 21 switches off the health variation in one channel at a time by assigning every state the value it takes in *GG*, and re-solves the decision problem in each case. Because the common-survivor mask and positive-earner samples are recomputed for

each paired experiment, the baseline variance for the channel exercises is 0.190, compared with 0.187 across the mandates. The difference reflects the experiment-specific comparison samples, not a change in the definition of cumulative earnings or its variance. Because the channels enter the Bellman equation, each exercise is a full re-solution rather than a reweighting, and the effects are not additive contributions to an accounting decomposition.

The utility cost of working is the largest single channel: removing its health variation lowers the variance of log cumulative earnings by 9.93%. The offered-earnings gradient accounts for 4.69%. Mortality has a negligible effect of -0.18% in this common-survivor comparison. The two employment-related channels overlap: removing both together gives 11.82%, short of the 14.62% their separate effects would sum to, because once health no longer shifts the cost of working, part of the earnings gradient it would otherwise have transmitted never materializes. Adding mortality to both leaves the figure at 11.83%.

These decomposition exercises have two implications. First, the model transmits health variation to the earnings distribution more strongly through employment participation than through offered earnings conditional on work, consistent with the substantially larger *BB* employment gap documented in Section 4. Second, the channel shutdowns and the mandates evaluate distinct mechanisms. A channel shutdown holds health dynamics fixed while eliminating state variation in one economic channel. A mandate retains those channels, changes the distribution of health states through the maintained transition process, and, when it requires healthy consumption, changes the resources individuals need. The similar magnitudes of the universal mandate and the employment-cost shutdown therefore do not imply that the mandate operates by equalizing the health-dependent cost of working.

Figure 21: Health channels and lifetime-earnings inequality



Notes: One health channel at a time is set to its *GG* value. Bars report the percentage change in the variance of log cumulative earnings over ages 25–55, relative to a baseline variance of 0.190. Negative values indicate that removing the channel’s health variation compresses the distribution. Combinations are reported in Appendix E.

7 Conclusion

The empirical evidence supports treating mental and physical health as distinct dimensions of the health–earnings relationship. Mixed health states have different associations with employment and earnings, and persistent health types capture substantial heterogeneity without eliminating transitions within type. Healthy choices are more common at higher incomes and strongly persistent, and much of their income gradient reflects who was already making them. Their conditional predictive associations with subsequent health differ across physical and mental outcomes.

The quantitative model connects these patterns through employment, saving, health transitions, costs of changing healthy choices, and permanent productivity. Persistence matters for the interpretation of the income gradient: once adopting a healthy choice is a commitment over several years, a moderate monetary cost of healthy consumption, about £2,900 a year, reproduces the gradient that a model without persistence could generate only with an implausibly large price.

The counterfactuals show that improving population health and compressing cumulative earnings are related but distinct objectives. Universal healthy choices deliver the largest aggregate health improvement in the reported battery and compress cumulative earnings substantially. Mandates targeted at low offered earnings compress cumulative earnings further, although they improve health by less. Lowering the price of healthy consumption improves health modestly and slightly increases dispersion by weakening the need to work. The size and incidence of gains, the resources required by an intervention, and the way decisions respond to it all matter for this comparison.

These results carry a lesson for the design of policies that promote healthy behaviors. Such policies are usually evaluated by their effect on population health, but their distributional consequences depend on whom they reach. An intervention that raises healthy choices across the whole population delivers the largest health gains while leaving much of its distributional potential unused, and a price reduction available to everyone can even widen earnings differences. When reducing inequality is an objective, the decisive margin is targeting: interventions that reach individuals with low earnings capacity, or the joint mental and physical health deficits concentrated among them, compress cumulative earnings far more per person covered than untargeted interventions. Policies that promote healthy behaviors and policies that reduce inequality can therefore reinforce each other, but only if healthy-behavior interventions are directed toward the individuals for whom poor health and low earnings coincide.

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A Measurement, samples, and descriptive regressions

A.1 Sample construction

Table A.1 records the number of person-years and distinct individuals remaining after each selection step, in the order the steps are applied. The funnel is cumulative, so the intermediate rows depend on that order; the final row is what the rest of the paper uses.

Table A.1: Sample selection

Step	Person-years	Individuals
Merged panel, waves 1–12	933,102	128,017
Non-missing health scores, longitudinal weight, age	207,738	36,794
Employment conditions (hours, earnings, real hourly wage)	182,773	34,726
Top 1% real-wage trim	181,931	34,678
Ages 25–76	145,568	26,853
Duplicate person-age records removed	140,881	26,853
Log-wage outliers	140,879	26,853
Estimation sample	140,879	26,853

The first step removes 78% of the merged panel and deserves an accounting, because it is the step at which a reader would suspect selection on health. It is driven almost entirely by longitudinal-weight eligibility rather than by missing health scores. Among person-years aged 25–76, the SF-12 scores are present for between 74% and 97% of observations in every wave. A positive longitudinal weight is present for none of wave 1—longitudinal weights require at least two observed waves, so the first wave carries none by construction—and for a share that falls from 68% in wave 2 to 38% in wave 12. The estimation sample therefore spans waves 2–12, eleven waves rather than twelve.

The selection steps in Table A.1 apply the following rules. Observations with missing core health variables or nonpositive weights are dropped. Working hours and real hourly wages are restricted before annual labor earnings are constructed, duplicate individual–age records are removed, and the upper real-wage trim excludes the top one percent.

The hourly-wage restrictions use wages expressed in 2015 pounds. The lower support cutoff is £5.72 at every survey date, and the pooled top-one-percent trim is also calculated from the real hourly-wage distribution. This convention prevents inflation from making the effective lower cutoff progressively less restrictive or mechanically shifting the composition of the upper-tail trim across waves. The retained data also contain specific adjustments to sparse late-age earnings cells, including an age-75 *BG* log-earnings replacement based on neighboring ages. These adjustments are relevant to late-life figures; they do not enter the age-25–60 distributional validation shown in this draft. Annual labor earnings are constructed from monthly amounts by multiplying by twelve, with the corresponding log adjustment.

A.2 Module periodicity

The seven behavioral inputs are not collected every wave. Table A.2 records where each is fielded among person-years aged 25–76.

Table A.2: Wave-by-wave availability of the survey modules used

Module	Waves fielded	Coverage when fielded
SF-12 mental and physical scores	1–12	0.74–0.97
Vigorous activity, moderate activity, walking	7, 9, 11, 12	0.64–0.98
Fruit and vegetable consumption	2, 5, 7, 9, 11	0.85–0.99
Smoking status	6–12	0.84–0.99
Alcohol consumption	7, 9, 11, 12	0.48–0.75
Employment, labor earnings, hours	1–12	0.90–1.00

The consequence is structural rather than incidental. All seven inputs are observed together only in waves 7, 9, and 11, so f^B and f^C exist for 24,214 person-years, distributed as 9,713, 7,870, and 6,631 across those three waves. Every quantity involving a healthy choice—the choice shares targeted in estimation, the income gradients, and the health transition estimations of Figure 4—rests on those waves. Those estimations pair a choice observed in wave 7, 9, or 11 with health observed in the following wave, which is why their common sample is 19,566 observations from 9,613 individuals rather than the full panel. Health, employment, and earnings are available every wave and are not subject to this restriction.

A.3 Attrition

Individuals are observed for 5.2 waves on average and 4 at the median, out of a possible eleven. The distribution is U-shaped: 20% of individuals contribute a single person-year and 11% contribute the maximum of eleven. The unconditional probability that an individual present in one wave does not reappear in the next is 0.217.

Exit is not independent of the variables the model is about. A logit for non-reappearance in the following wave, weighted and clustered by individual, gives the average marginal effects in Table A.3. Relative to individuals in good health on both margins, those in poor health on both are 4.9 percentage points more likely to leave. On the subset of waves where the behavioral module is fielded, individuals making the healthy-consumption choice are 3.4 points less likely to leave, while the healthy-behavior choice is not distinguishable from zero.

Two qualifications bound what this establishes. First, non-reappearance mixes survey attrition with death, and death is a modeled process rather than a measurement problem: part of the elevated exit rate in BB is the mortality that the survival schedule $s(j, H)$ is estimated to capture. The logit cannot separate the two, so 4.9 points is an upper bound on the attrition component. Second, the direction of the remaining selection is known even though its magnitude is not: individuals in poor health on both margins and individuals not making the healthy-consumption choice leave faster, so the observed panel under-represents them at later ages, and estimates that condition on being observed will understate the prevalence of the states in which the paper’s mechanism operates.

Table A.3: Probability of not reappearing in the following wave

Average marginal effect	All waves (1)	Module waves (2)
<i>BB</i>	0.0485*** (0.0043)	0.0401*** (0.0100)
<i>BG</i>	0.0108*** (0.0039)	0.0119 (0.0087)
<i>GB</i>	0.0157*** (0.0036)	0.0074 (0.0081)
Healthy behavior	—	−0.0067 (0.0066)
Healthy consumption	—	−0.0344*** (0.0067)
Employed	−0.0037 (0.0037)	0.0319*** (0.0094)
Person-years	132,959	24,214

Notes: Average marginal effects from a weighted logit for non-reappearance in the following wave; standard errors clustered by individual in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The longitudinal weights used throughout are constructed by the survey to correct for differential non-response on observed characteristics, and they are the reason the descriptive moments and estimation targets are weighted rather than raw. They do not address selection on unobservables. A sharper treatment would require an explicit selection model estimated jointly with the structural parameters, which this paper does not attempt; the exercise above is a bound on how much room such a model would have to matter, not a demonstration that it would not.

A.4 Health and permanent types

The health classifications use the SF-12 mental and physical component scores separately. Weighted quintiles determine the binary thresholds, with the lower two quintiles classified as Bad. Combining the binary indicators yields the state coding 1 = *BB*, 2 = *BG*, 3 = *GB*, and 4 = *GG*.

Table A.4 reports the labor-market regressions used to choose the common cutoff. Each column comes from a separate weighted regression with score-quintile indicators, a cubic in age, sex, education, and year effects; the omitted category is the healthiest quintile. Employment is estimated on all eligible person-years and log earnings among employed workers. The largest penalties occur toward the bottom of each score distribution, especially for physical health. We use the lower two quintiles as a parsimonious common definition across both dimensions rather than estimate separate outcome-specific cutoffs. For the MCS, this bottom-40-percent rule is also close to the threshold in Jolivet and Postel-Vinay (2025), whose Severe and Poor categories together comprise the bottom 36 percent of their distribu-

tion.

Table A.4: SF-12 score quintiles and labor-market outcomes

Score quintile	Employment		Log earnings	
	MCS	PCS	MCS	PCS
Q1 (lowest)	−0.160	−0.257	−0.031	−0.156
Q2	−0.015	−0.072	−0.008	−0.100
Q3	0.022	−0.024	0.017	−0.062
Q4	0.032	−0.002	0.038	−0.017
Reference	Q5	Q5	Q5	Q5

Notes: Coefficients from separate survey-weighted regressions. All specifications include a cubic in age, sex, education, and year effects. Log-earnings regressions condition on employment.

Permanent health types are constructed from individual means of residualized mental and physical scores. The residual regressions remove a quadratic age profile, sex, and education. Four-cluster k -means, initialized with the fixed seed 6666 in the data pipeline, groups the resulting pairs of individual means. Labels are aligned across outputs as type 1 (relatively good in both dimensions), type 2 (relatively poor in both), type 3 (predominantly poor physical health), and type 4 (predominantly poor mental health). A type label describes the cluster, not a restriction that its members always occupy the corresponding current health state.

Using the available health history improves the signal about the persistent component of individual health, although the precision of the classification depends on the number of observations available for each person. The assigned type is a generated, reduced-form measure that reflects both permanent biological components and persistence in observed health trajectories. The displayed regression intervals condition on the constructed types rather than reestimating the complete classification in each replication.

A.5 Healthy-choice construction and cleaning

The seven inputs are vigorous activity, moderate activity, walking, fruit consumption, vegetable consumption, smoking status, and heavy drinking. Weighted principal-component analysis retains two components, followed by varimax rotation. Table A.5 records the saved loadings used by the figure pipeline. Component signs and variable coding must be preserved when reproducing the scores; changing a sign would also change the interpretation of the zero threshold. Zero is the origin of the score produced by projecting the standardized inputs onto each rotated component. A score above zero lies in the direction labeled healthy for that component, while a score below zero lies on the opposite side; it is not a median split or a count of healthy actions.

Table A.5: Saved rotated component loadings

Input	Consumption	Behavior
Vigorous activity	0.007	0.619
Moderate activity	-0.003	0.635
Walking	0.061	0.422
Fruit consumption	0.585	0.039
Vegetable consumption	0.546	0.078
Smoking-status variable	0.441	-0.078
Heavy-drinking variable	-0.401	0.149

A.6 Separate binary logits in Figure 4

For outcome $q \in \{m, p\}$, let G_{it}^q indicate good health. The full specification is

$$\Pr(G_{it}^q = 1 \mid X_{it}) = \Lambda(\alpha_q + \beta_C^q f_{i,t-1}^C + \beta_B^q f_{i,t-1}^B + D'_{it} \delta_q + \kappa_m^q G_{i,t-1}^m + \kappa_p^q G_{i,t-1}^p + \xi_{\eta_i}^q), \quad (11)$$

where Λ is the logistic cdf and D includes age, age squared, year indicators, and a sex indicator. Specification 1 omits lagged health and type indicators; specification 2 adds the two lagged binary health indicators; specification 3 adds types. There are no lagged composite-state interaction indicators in these specifications.

The six logits use the same 19,566 consecutive-wave observations and longitudinal survey weights. For a binary choice f , the displayed average discrete effect is

$$100 \frac{\sum_{it} w_{it} \{\hat{P}_{it}(f=1) - \hat{P}_{it}(f=0)\}}{\sum_{it} w_{it}},$$

with other regressors held at their observed values. Thus, the plotted quantities are percentage-point probability changes, not raw log-odds coefficients. Confidence intervals use the delta method and individual-clustered coefficient covariance. The weighting adjusts the contribution of observations; clustering adjusts their estimated uncertainty.

A.7 Income definitions across the empirical and counterfactual analyses

Figure 7 uses the pooled weighted median of real gross monthly income among eligible person-years aged 25–75 with both choice indicators observed. Its reported difference is the lower-income share minus the higher-income share within age group. Observations exactly at the median are excluded from that comparison.

The structural income moments instead use positive labor earnings at ages 25–60 and residualize log earnings on quadratic age and the joint binary health indicators. They divide the sample at a weighted residual-income median and construct four weighted choice shares. The data targets are 0.473768 and 0.469252 for behavior above and below the median, and 0.591065 and 0.510119 for consumption. Table 3 uses this second definition. Comparing the

sizes of the gradients across Figure 7 and Table 3 without accounting for sample, income concept, and residualization would be misleading.

The income-contingent counterfactuals use a third definition. They calculate P10, P25, and P50 separately at every age from positive annual labor earnings in the baseline simulation, without residualizing earnings. The resulting monetary schedules are held fixed in the counterfactual, and eligibility compares them with offered annual earnings, whether or not the individual works. Thus, neither the pooled gross-income split in Figure 7 nor the residual-income split targeted in Table 3 determines counterfactual eligibility.

B External identification of the earnings process

B.1 Deterministic earnings component

The deterministic component of earnings is estimated by weighted least squares among employed individuals aged 25–60 with positive labor earnings,

$$\log y_{it} = \beta_0 + \beta_1 \text{age}_{it} + \beta_2 \text{age}_{it}^2 + \beta_3 \text{age}_{it}^3 + \sum_{h \in \{BB, BG, GB\}} \lambda_h \mathbf{1}\{H_{it} = h\} + X'_{it}\delta + u_{it},$$

where y_{it} is annual labor earnings in 2015 pounds, H_{it} is composite health with GG as the reference, and X_{it} contains indicators for urban residence, race, marital status, sex, education, and survey year. Observations are weighted by the longitudinal survey weight. The age coefficients define the lifecycle profile $g(24+j) = \beta_1(24+j) + \beta_2(24+j)^2 + \beta_3(24+j)^3$ used in equation (3), and the health coefficients define λ_h . Table B.1 reports the estimates.

Table B.1: Deterministic component of log annual labor earnings

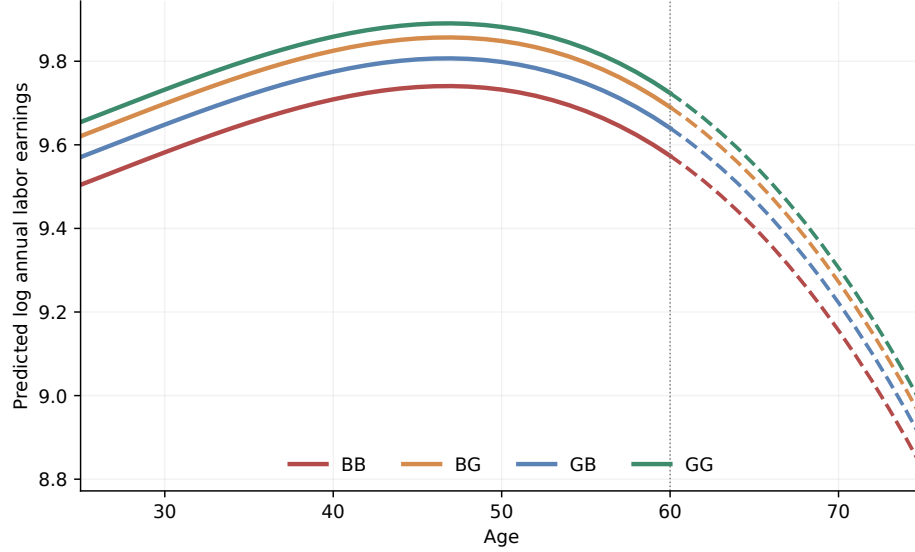
	Estimate	Standard error
Age	−0.01232	0.02200
Age ²	0.001045	0.000531
Age ³ ($\times 10^3$)	−0.01302	0.00414
<i>BB</i>	−0.1500	0.0110
<i>BG</i>	−0.0337	0.0074
<i>GB</i>	−0.0838	0.0085
Constant	9.577	0.296
Observations		71,793
Individuals		15,645
R^2		0.244

Notes: Weighted least squares among employed individuals aged 25–60. The regression also includes indicators for urban residence, race, marital status, sex, education, and survey year. Health coefficients are relative to GG . Standard errors are clustered by individual.

Joint deficits carry the largest earnings penalty: relative to GG , earnings are 15.0 log points lower in BB , 8.4 points lower in GB , and 3.4 points lower in BG . Figure 22 shows

the implied lifecycle profile of log earnings in each health state, evaluated at the weighted sample means of the remaining controls. The profile peaks at about age 47, and the health differentials shift it in parallel. The model uses the cubic over the whole horizon; beyond age 60 the profile is an extrapolation of the estimated polynomial.

Figure 22: Predicted lifecycle profile of log labor earnings by composite health state



Notes: Predicted log annual labor earnings from the regression in Table B.1, with the remaining controls at their weighted sample means. Solid lines cover the estimation range, ages 25–60; dashed lines extrapolate the cubic to age 75, as in the model.

B.2 Stochastic earnings component

The alternative earnings-process code estimates residuals only within the employed, positive-earnings sample aged 25–60. Its first-stage regression contains cubic age, composite health, urban status, race, marital status, sex, education, and year controls, using survey weights. Residuals are retained only for observations in the regression estimation sample. A covariance at lag k requires that both endpoints are eligible and separated by k annual periods. Panel lags respect gaps; eligibility of every intermediate observation is not required.

Under the decomposition $r_{it} = a_i + z_{it} + e_{it}$, mutual orthogonality and stationarity of z imply equation (9). For positive lags,

$$C(k) - C(k+1) = B\rho^k(1 - \rho), \quad \frac{C(k+1) - C(k+2)}{C(k) - C(k+1)} = \rho.$$

This illustrates the source of identification: the rate at which positive-lag covariances decline identifies persistence; their persistent level helps separate A from B ; and the excess of $C(0)$ above $A + B$ identifies E . The implementation fits all included covariances jointly rather than estimating these potentially noisy ratios one at a time.

The estimator minimizes equally weighted squared discrepancies for lags zero through eight. For each candidate persistence, an active-set least-squares calculation imposes non-negative variance components. A grid over $\rho \in [0.001, 0.9999]$ is followed by local refinement.

Covariance construction uses current-endpoint weights. The alternative-analysis code also provides comparisons with specifications omitting permanent or independent components and sensitivity to shorter maximum lags. Its individual-cluster bootstrap uses 499 replications, seed 170912, and reestimates the first-stage regression. The point estimates used by this draft are in Table B.2. These external estimates are treated as known when the structural parameters are estimated, so the structural standard errors reported in Table D.2 condition on them and do not propagate the uncertainty of this first stage.

Table B.2: External residual covariance decomposition and model use

Parameter	Estimate	Role in current model
Persistence ρ	0.872167	Imported
Innovation variance σ_ν^2	0.021291	Imported
Innovation standard deviation σ_ν	0.145913	Implied simulation input
Stationary AR variance B	0.088961	Implied by imported process
Permanent residual variance A	0.071676	External identification only
Independent residual variance E	0.024390	Excluded from simulation

Allowing $A > 0$ matters because a slowly decaying covariance need not be generated entirely by a highly persistent shock. Conversely, in a short panel a permanent component and a very persistent autoregressive component can be difficult to separate. The decomposition should be judged by lag sensitivity and uncertainty, not simply by its extra flexibility. Selection also remains relevant: these are covariances of earnings observed among workers, whereas the model requires a process for offered earnings. The external regression does not implement a separate selection correction.

There are two further distinctions between this decomposition and the structural simulation. First, the external covariance equation uses a stationary AR component, whereas the structural model starts with $z_1 = 0$. We import its law of motion, not its initial stationary distribution. Second, A is not automatically equal to any $\sigma_{\theta,k}^2$. The residual regression removes a different set of observables, and the structural distribution is conditional on health type and identified jointly with employment. Adding an independent variance A to that distribution would risk counting permanent heterogeneity twice.

The model initializes the persistent component at $z_1 = 0$. The untruncated autoregressive process then implies

$$\text{Var}(z_j) = \sigma_\nu^2 \frac{1 - \rho^{2(j-1)}}{1 - \rho^2}.$$

This expression helps explain why earnings risk accumulates rapidly and then flattens. It is not a formula for the observed worker variance: permanent productivity, health, covariances among these components, endogenous selection, and numerical shock bounds also matter. Matching the worker variance through all these channels is more demanding than reproducing the variance of the exogenous shock alone.

C Health transitions and their recalibration

The original Stata specification is a multinomial logit for next year’s composite state, with GG as the base outcome. Its regressors are current composite-state indicators, both current healthy choices, age, and permanent health-type indicators, weighted by `avg_wght`. This joint model permits dependence between next period’s mental and physical health. The two binary logits in Figure 4 serve a separate descriptive purpose and do not generate the transition matrix.

Let $P_{hh'}(f)$ denote a row of original predictions at fixed age, type, and choice bundle $f = (f^B, f^C)$. The transition matrix used for the reported structural estimates applies two probability-level adjustments. First, origin-specific parameters change retention:

$$P'_{hh} = \frac{P_{hh} + \lambda_h^R}{1 + \lambda_h^R}, \quad P'_{hh'} = \frac{P_{hh'}}{1 + \lambda_h^R} \quad (h' \neq h).$$

Second, destination parameters redistribute the off-diagonal mass while preserving the adjusted diagonal:

$$P''_{hh} = P'_{hh}, \quad P''_{hh'} = (1 - P'_{hh}) \frac{P'_{hh'} \exp(\mu_{h'}^R)}{\sum_{k \neq h} P'_{hk} \exp(\mu_k^R)}, \quad h' \neq h.$$

The fitted retention adjustments are $\lambda_{BB}^R = 0.471$, $\lambda_{BG}^R = 0.019$, $\lambda_{GB}^R = 0.271$, and $\lambda_{GG}^R = 0.332$. The destination adjustments are $\mu_{BB}^R = -0.639$, $\mu_{BG}^R = -0.505$, and $\mu_{GB}^R = 0.188$, with $\mu_{GG}^R = 0$. The retention parameters are fitted to departure hazards and the destination parameters to age-specific health shares, using empirical choice propensities and survival in the fitting simulation.

Because probability-level transformations need not preserve the choice-related log-odds contrasts inherited from the multinomial logit, we also implement a choice-preserving recalibration as an audit and candidate replacement. It separates the correction of aggregate health dynamics from the empirical choice coefficients. The structural parameters and headline counterfactuals must be re-estimated before this alternative replaces the matrix underlying the reported quantitative results.

For each row, define $q_{hh'}(f) = \log P_{hh'}(f)$. We add an origin-specific stay shift λ_h^R and a destination shift $\mu_{h'}^R$:

$$\tilde{q}_{hh'}(f) = q_{hh'}(f) + \lambda_h^R \mathbf{1}\{h' = h\} + \mu_{h'}^R,$$

and recover probabilities using the softmax

$$\tilde{P}_{hh'}(f) = \frac{\exp\{\tilde{q}_{hh'}(f)\}}{\sum_k \exp\{\tilde{q}_{hk}(f)\}}.$$

The superscript R distinguishes these parameters from the earnings health gradient, and $\mu_{GG}^R = 0$ normalizes the destination indices. Because neither shift varies with f , this alternative preserves exactly the choice-related log-odds contrasts:

$$\left[\log \frac{\tilde{P}_{hh'}(f)}{\tilde{P}_{h,GG}(f)} - \log \frac{\tilde{P}_{hh'}(0)}{\tilde{P}_{h,GG}(0)} \right] = \left[\log \frac{P_{hh'}(f)}{P_{h,GG}(f)} - \log \frac{P_{hh'}(0)}{P_{h,GG}(0)} \right].$$

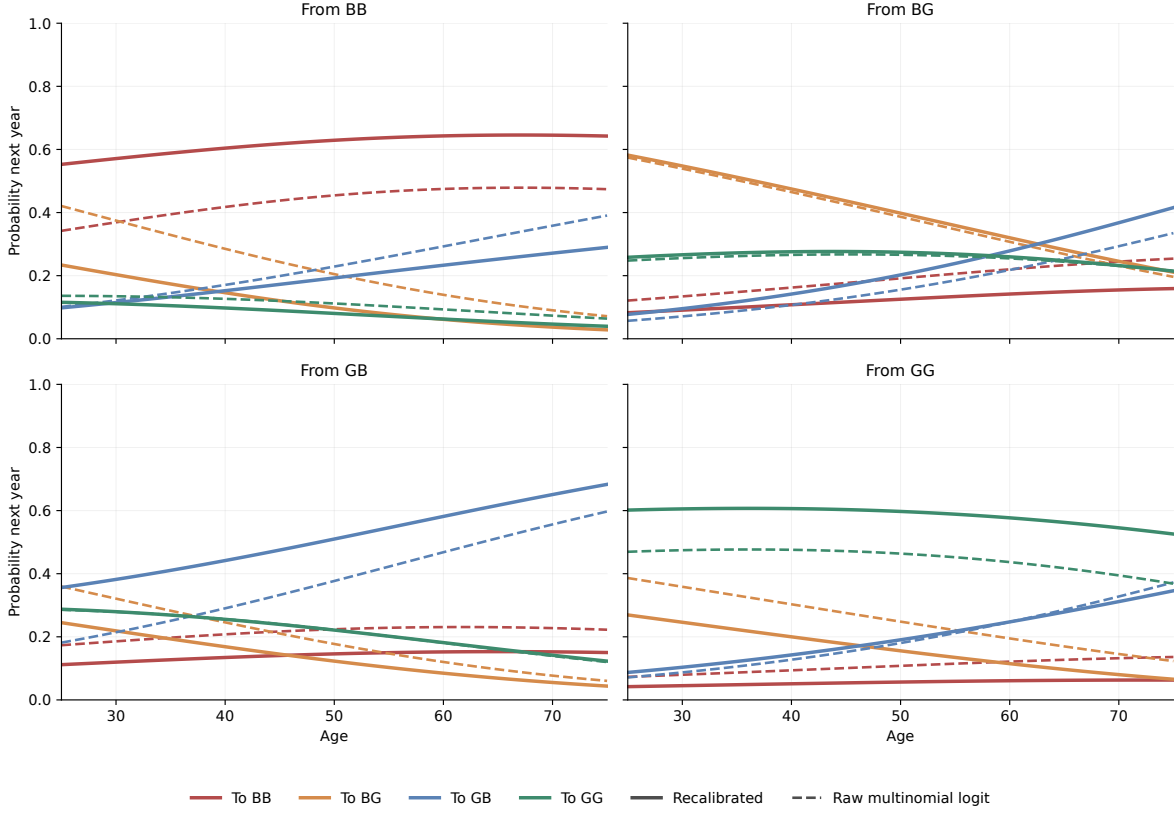
Thus the alternative adjustment changes baseline persistence and destination incidence without redefining the estimated association between healthy choices and subsequent health.

The four stay shifts are first fitted to observed departure hazards. Holding them fixed, the three free destination shifts are fitted to age-specific health shares. Relative to the raw transition matrix, the sum of squared errors for age-specific health shares falls from 0.4469 to 0.0490 and that for spell-departure hazards falls from 0.1412 to 0.0665. The corresponding errors under the reported probability-level recalibration are 0.0603 and 0.0752. Its largest distortion of a choice-related log-odds contrast is 0.419, whereas the largest numerical change under the index-shift alternative is 2.2×10^{-15} , at machine precision. These diagnostics assess the two targeted margins; they do not provide independent causal validation of the observational choice coefficients.

C.1 Raw and recalibrated transitions

Figure 23 compares the one-year transition probabilities of the original multinomial logit with those of the recalibrated matrix used in the model. Each panel fixes the origin state and shows the probability of each destination over age, for individuals who make neither healthy choice, averaging permanent types with their distribution at age 25. The recalibration changes the matrix in two ways. First, it raises persistence in three of the four states. At age 45, the probability of remaining in *BB* rises from 0.44 to 0.62, in *GB* from 0.33 to 0.47, and in *GG* from 0.47 to 0.60, while retention in *BG* is almost unchanged (0.43 to 0.44). The raw logit therefore implies too much movement across states relative to the observed durations of health spells. Second, it redirects departures away from *BB* and *BG* and toward *GB*. At age 45, the probability of moving from *GG* to *BG* falls from 0.28 to 0.18 and from *BB* to *BG* from 0.24 to 0.12, whereas the probability of moving from *BG* to *GB* rises from 0.13 to 0.17. Without this adjustment, the raw matrix places too many person-years in *BG* and too few in *GB* and *GG*: the sum of squared errors for age-specific health shares falls from 0.4469 with the raw matrix to 0.0603 with the recalibrated one. The cost is a worse fit of some one-year flows, in particular transitions into *BG*, which the destination adjustment reduces to match the stock of *BG*.

Figure 23: Raw and recalibrated health transitions



Notes: One-year transition probabilities by origin state and age for individuals who make neither healthy choice, averaging permanent types with their distribution at age 25. Dashed lines show the original multinomial-logit predictions and solid lines the recalibrated matrix used in the model.

C.2 Survival probabilities

Mortality is first estimated with an unweighted logit for death before the next wave,

$$\Pr(D_{it} = 1 \mid \text{age}_{it}, H_{it}) = \Lambda(a_0 + a_1 \text{age}_{it} + c_{H_{it}}), \quad c_{GG} = 0,$$

where D_{it} records a death outcome at the following interview. The estimates are $a_0 = -11.71$, $a_1 = 0.0835$, $c_{BB} = 2.077$, $c_{GB} = 1.228$, and $c_{BG} = 0.235$: conditional on age, the odds of death are about eight times higher in BB than in GG . Because deaths are observed only among sample members who remain in contact with the survey, the implied mortality is below population mortality. We therefore shift the log-odds of death by an age-specific constant $\delta(\text{age})$, common to the four health states:

$$q_h(\text{age}) = \Lambda(a_0 + a_1 \text{age} + c_h + \delta(\text{age})), \quad \sum_h \omega_h(\text{age}) q_h(\text{age}) = q^{\text{ONS}}(\text{age}).$$

Here $\omega_h(\text{age})$ is the share of composite health state h at each age in the UKHLS, and $q^{\text{ONS}}(\text{age})$ is the one-year mortality rate in the national life tables of the Office for National Statistics for the United Kingdom, averaged over the periods 2009–2011 to 2019–2021 and

over men and women using the sex composition of the UKHLS at each age. The shift $\delta(\text{age})$ is positive at every age, falling from 1.04 at age 25 to 0.45 at age 55 and rising to 0.59 at age 75. The resulting annual death probabilities at age 65 are 0.024 in *BB*, 0.010 in *GB*, 0.004 in *BG*, and 0.003 in *GG*. The model uses $s_j(h) = 1 - q_h(24 + j)$.

D Simulated moments, numerical solution, and parameters

Table D.1 states the structural moment blocks. Each moment enters the criterion with the weight implied by $\hat{\Omega}^{-1}$. The health-share and retention diagnostics are outside this 145-moment list, although they inform the preceding health recalibration. Similarly, earnings autocorrelations and the full age profiles of conditional mean earnings are diagnostic figures rather than one-for-one targets in this criterion.

Table D.1: Moment blocks in structural estimation

Indices	Moments	Count
1–40	Employment by health and five-year age bin	40
41–80	Healthy behavior, same cells	40
81–120	Healthy consumption, same cells	40
121–124	Two choices above/below residual-income median	4
125–137	Early-adult earnings levels and dispersion	13
138–141	Healthy choices given the previous choice (two-year transitions)	4
142–145	Healthy consumption by residual-income half and previous choice	4

The last block contains four mean log-earnings moments by health type at ages 25–29, four corresponding variances, and five pooled variances at individual ages 25 through 29. A match to that block is not a match to the entire lifecycle variance profile. It nevertheless means that an “untargeted variance” label must be restricted to later ages. The last two blocks use the person-years at ages 27–60 whose previous observation of the behavioral module lies one to three years earlier; because the module is fielded every other wave, about 92% of these pairs are exactly two years apart. The model counterpart compares each choice with the choice made two years earlier.

Optimization and the first-order condition

The criterion is minimized by Levenberg–Marquardt on the weighted residuals, using a finite-difference Jacobian $\hat{G}$ built with common random numbers: all $2K$ perturbed evaluations share the simulation seed, so the differences reflect the parameter change rather than simulation noise. The step for each coordinate is proportional to that coordinate’s magnitude (0.5 percent) rather than set to machine precision, because several moments are computed from realized employment draws and are therefore step functions of ϑ at a fixed seed; a machine-precision step returns quantization noise rather than a derivative. Proportional steps matter because the utility-cost coefficients differ by orders of magnitude: the age-squared terms of

the healthy-choice costs are of order 10^{-7} to 10^{-6} , and a common absolute step would move utility by more than the scale of the taste shocks.

A gradient-based method matters here because the criterion is informative about coordinates that a derivative-free search can leave untouched. The nine age-by-health interaction coefficients of the employment cost are a case in point: their Jacobian block has full rank, each moves the employment cells of its own health state, and their column norms are comparable to those of the coefficients that identify the reference profile. Estimating them is what closes the late-career employment discrepancies.

For the reported inference calculation, the Jacobian is evaluated with 400,000 simulated lifecycles, considerably more than the sample used during optimization. This reduces simulation noise in the local curvature calculation and permits direct estimation of the simulation covariance entering the sandwich formula below. Because the coefficients are measured in very different units, from 10^{-7} for the age-squared cost terms to thousands of pounds for p_f , the condition number of $\hat{A}$ in the original units mainly reflects scale. Measured scale-free, the weighted Jacobian has full column rank, and the correlation matrix of the estimates has a condition number of about 10^4 . The remaining ill-conditioning comes from coefficients that partly substitute for one another: the linear and quadratic age terms of each healthy-choice cost (correlation -0.98 for healthy behavior and -0.90 for healthy consumption), the quadratic and cubic age-by-health interactions of the employment cost, and the switching costs and the taste-shock scale, since utility is identified only relative to τ and the data pin down κ_B/τ and κ_C/τ more precisely than either component. The reported standard errors should therefore be interpreted as local uncertainty measures; individual polynomial coefficients can be imprecise even when the age profile they jointly generate is well determined.

The estimation configuration uses 100 asset nodes, seven persistent-shock nodes, nine permanent-productivity nodes, four health types, four current health states, and four combinations of previous healthy choices, and simulates 50,000 lifecycles per criterion evaluation. The independent-shock grid has one node because its variance is zero. The asset domain runs from 0 to 1,000,000; initial assets are 21,085. The initial value is the mean net wealth observed for ages 20–24 and serves as an estimate of wealth upon entry at age 25; the lower bound rules out borrowing. The moment covariance is not constructed from the simulation; it is the individual-cluster bootstrap of the data moments described below. The solver applies backward induction, an endogenous-grid calculation conditional on each discrete bundle, and interpolation along the asset dimension. Forward simulation draws choices, health transitions, earnings innovations, and survival; conditional choice probabilities are used for selected share moments to reduce simulation noise, while realized histories determine employment and cumulative earnings. The persistent process is discretized for expectations; simulated innovations are bounded at the shock-grid limits.

Table D.2 reports every estimated structural parameter together with its sandwich standard error. Let $\hat{G}$ denote the common-random-number finite-difference Jacobian of the simulated moments and $\hat{S}$ the simulation covariance estimated from 300 independent simulation replications. The reported covariance is

$$\hat{V} = \hat{A}^{-1} \hat{G}' \hat{\Omega}^{-1} (\hat{\Omega} + \hat{S}) \hat{\Omega}^{-1} \hat{G} \hat{A}^{-1}, \quad \hat{A} = \hat{G}' \hat{\Omega}^{-1} \hat{G}.$$

The Jacobian uses 400,000 simulated lifecycles. With $M = 145$ moments and $K = 41$ free coefficients, the model has 104 overidentifying restrictions. Standard errors are reported in

parameter levels; the delta method is applied to the log-parameterized choice-shock scale and type-specific productivity standard deviations. They condition on the externally estimated health, survival, and earnings processes. Type-specific means are levels of log productivity and their standard deviations are not variances. Fixed parameters, including b , receive no structural standard error.

Table D.2: Structural parameter estimates and sandwich standard errors

Parameter	Estimate	Sandwich s.e.
<i>Healthy-behavior cost schedule</i>		
ψ_0^B	-0.001281***	0.000423
ψ_1^B	2.99×10^{-7}	2.43×10^{-5}
ψ_2^B	2.58×10^{-7}	8.11×10^{-7}
Δ_{BB}^B	0.003605***	0.000389
Δ_{BG}^B	0.001475***	0.000272
Δ_{GB}^B	0.002402***	0.000334
<i>Healthy-consumption cost schedule</i>		
ψ_0^C	-0.002509***	0.000208
ψ_1^C	0.000115***	1.49×10^{-5}
ψ_2^C	-3.69×10^{-6} ***	5.08×10^{-7}
Δ_{BB}^C	0.002619***	0.000306
Δ_{BG}^C	0.002429***	0.000355
Δ_{GB}^C	0.002221***	0.000356
<i>Participation-cost schedule</i>		
ψ_0^n	-0.009838***	0.000949
ψ_1^n	-0.001994	0.003641
ψ_2^n	-0.129206***	0.010768
ψ_3^n	0.218094***	0.015184
Δ_{BB}^n	0.011961***	0.001369
Δ_{BG}^n	0.003755***	0.000852
Δ_{GB}^n	0.006398***	0.001187
$\psi_{n,1}^{BB}$	0.050698***	0.007835
$\psi_{n,1}^{BG}$	0.017743***	0.005682
$\psi_{n,1}^{GB}$	0.007285	0.006118
$\psi_{n,2}^{BB}$	-0.075155***	0.014099
$\psi_{n,2}^{BG}$	-0.060811***	0.013692
$\psi_{n,2}^{GB}$	-0.019468	0.013757
$\psi_{n,3}^{BB}$	0.011094	0.008839
$\psi_{n,3}^{BG}$	0.037112***	0.010931
$\psi_{n,3}^{GB}$	0.004643	0.010958
<i>Switching costs</i>		
κ_B	0.013068***	0.000950
κ_C	0.017555***	0.001290

Table D.2 (continued)

Parameter	Estimate	Sandwich s.e.
<i>Preferences, prices, and permanent productivity</i>		
p_f (£)	2938.56***	364.84
ϕ_B	0.405440***	0.114555
τ	0.009669***	0.000660
$\mu_{\theta,1}$	9.578939***	0.013345
$\mu_{\theta,2}$	9.448082***	0.048592
$\mu_{\theta,3}$	9.543655***	0.021479
$\mu_{\theta,4}$	9.487383***	0.019231
$\sigma_{\theta,1}$	0.362569***	0.010408
$\sigma_{\theta,2}$	0.308730***	0.037837
$\sigma_{\theta,3}$	0.331770***	0.018398
$\sigma_{\theta,4}$	0.393457***	0.013420
<i>Fixed parameters</i>		
β	0.960000	—
r	0.030000	—
α	0.600000	—
γ	2.000000	—
b	0.011000	—

Notes: Sandwich standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, two-sided tests based on the normal approximation. Fixed parameters are not estimated.

For employment, the estimated baseline cost as a function of $t = (j - 1)/50$ is

$$\psi_j^n(GG) = -0.009838 - 0.001994t - 0.129206t^2 + 0.218094t^3.$$

The health intercepts $(\Delta_{BB}^n, \Delta_{BG}^n, \Delta_{GB}^n)$ are (0.011961, 0.003755, 0.006398). Ordered in the same way, the interaction coefficients $\psi_{n,m}^h$ for the linear, quadratic, and cubic age terms are

$$\begin{aligned} \text{linear : } & (0.050698, 0.017743, 0.007285), \\ \text{quadratic : } & (-0.075155, -0.060811, -0.019468), \\ \text{cubic : } & (0.011094, 0.037112, 0.004643). \end{aligned}$$

The signs alternate across the three polynomial terms within each health state, so the profiles they generate are not monotone shifts of the reference profile but differences in the age at which the cost of working rises steeply. This is what allows employment to fall at different ages depending on which health margin has failed.

The weighting matrix

$\hat{\Omega}$ is computed by an individual-cluster bootstrap on the estimation sample. Each of $B = 2,000$ replicates draws $n = 26,853$ individuals with replacement, carrying each individual's

entire observed history, and recomputes all 145 moments with the same code path that produces the point estimates. The replicate covariance is $\hat{\Omega}$. Held fixed across replicates, and therefore conditioned on, are three objects estimated upstream on the same sample: the permanent health types (k-means on residualized health histories), the composite health-state cutpoints (weighted quintiles), and the two behavior indices (weighted principal components with a varimax rotation, thresholded at zero). Recomputing them per replicate invites label switching across replicates, and we prefer a stated conditioning to an unstable one; the cost is that $\hat{\Omega}$ understates sampling variance to the extent those first-stage objects are themselves estimated, most plausibly for the moments that condition on permanent type.

B is set by conditioning rather than by rank. Rank requires only $B > M$. The binding constraint is that a bootstrap covariance with ratio $q = M/B$ carries a spurious condition number even when the population correlation matrix is well conditioned: the Marčenko–Pastur edges $[(1 - \sqrt{q})^2, (1 + \sqrt{q})^2]$ imply a factor of roughly 28 at $B = 300$ and 2.9 at $B = 2,000$. At the chosen B the correlation matrix of $\hat{\Omega}$ has condition number 320, so no shrinkage or ridge is applied and the matrix enters (10) as estimated. No moment required a variance floor: the smallest implied design effect across the 120 share moments is 1.13, comfortably above the independent-sampling benchmark that would signal a degenerate cell.

The two smallest eigendirections of the correlation matrix are structural rather than artifacts of finite B , and more replicates would not remove them. The smallest loads the type-specific early-career earnings variances against the age-specific ones; these are two marginals of the same table computed on the same person-years. The next ones load the income-gradient choice shares against the behavior and consumption cells they are computed from, and the unconditional income-gradient shares against their counterparts conditional on the previous choice.

Why working carries a utility cost and not a time cost

Working is costly in the model only through $\psi_j^n(H)$, which enters utility additively. An earlier specification instead made it costly in time: working consumed a health- and age-dependent fraction $\phi^\ell(j, H)$ of the unit time endowment, so that leisure was $1 - \phi^\ell(j, H) n - f^B \phi_{fB}$, with ϕ^ℓ a cubic in age carrying its own health intercepts and age-by-health interactions—sixteen parameters with the same functional form $\psi_j^n(H)$ has now. We report why that specification was abandoned, because it bears on how the participation profile should be read.

The two costs are hard to tell apart. Both act on the same margin and are identified from the same moments, the employment shares by age and health state, and the estimator put essentially all of the health-specific variation in one of them. The health intercepts of ϕ^ℓ came out at (0.0007, 0.0018, 0.0001) for BB , BG , and GB , and the nine age-by-health interactions at order 10^{-4} . Those twelve coefficients exist to let the timing of withdrawal from work differ by health, and they did not: the implied time cost crossed the entire time endowment at age 67 in all four health states, to three decimal places. The health-specific content of the estimate was carried by the utility cost, and the time cost contributed a common age profile that the utility cost could equally well have produced.

A second problem is that the time cost does more than raise the price of working. When $\phi^\ell(j, H)$ reaches one, leisure is nonpositive, the consumption–leisure aggregate is undefined,

and working leaves the choice set: participation is not small but exactly zero, and no wage restores the option. That cutoff is a discontinuity rather than a steep slope, and it overshoots. Under that specification, model employment among individuals in good health at ages 70–74 was 0.000 against 0.035 in the data, while the data still show people working at those ages.

Dropping the time cost therefore removes a parameter block that was not separately identified and a feature that the data reject, at the price of requiring the remaining utility cost to generate the entire participation profile, including its late decline. A quadratic ψ^n could not: steepening it enough to bring late-career employment down pushed the profile at ages 50–59 below the data. The cubic form with age-by-health interactions reported above is what replaced it. Unlike the interactions of ϕ^ℓ , its interactions are separately identified, and those for BB and BG are mostly estimated away from zero; the GB interactions are small and imprecise.

E Counterfactual definitions and aggregation

For scenario q , define cumulative earnings

$$Y_i^q = \sum_{a=25}^{55} n_{ia}^q w_{ia}^q \mathbf{1}\{i \text{ alive at } a \text{ in } q\}.$$

The comparison first restricts to individuals alive at age 55 in both baseline and counterfactual. The log-variance statistic then excludes nonpositive cumulative earnings within each scenario. Thus, the survival sample is common but the positive-earner samples need not be identical if an intervention changes whether someone ever works. Mortality effects outside this common sample and changes in zero cumulative earnings are not summarized by this variance alone.

Each intervention is compared with its own paired baseline, using the same initial draws and individual shock streams. Because the common-survivor mask differs slightly across interventions, so does the baseline statistic each is divided by; we report $100(V^{cf}/V^{base} - 1)$ against each experiment’s own denominator rather than against a single pooled one. Health shares instead aggregate all alive person-years at ages 25–75, including both treated and untreated groups; they are not restricted to the common-survivor sample used for earnings. The health-state bars in Figure 19 report percentage-point differences, while Figure 18 reports relative percent changes.

Health-contingent interventions impose both choices, behavior alone, or consumption alone in BB ; another imposes both choices whenever health is not GG . Mandates restrict the choice set but do not change prices: an individual required to make the healthy-consumption choice pays p_f from current resources, and the no-borrowing constraint and terminal solvency condition continue to apply. Individuals for whom no admissible path satisfies these constraints are excluded from both the baseline and the counterfactual simulation, as described in Section 6. Universal imposition applies both choices across health states. For each earnings-targeted rule, we compute a separate monetary cutoff at every age from the selected percentile of positive contemporaneous labor earnings in the baseline simulation and hold that schedule fixed in the counterfactual. Eligibility compares the cutoff with offered

annual earnings, which exist whether or not the individual works. In the baseline, 9.96%, 25.18%, and 50.14% of alive person-years are eligible under the P10, P25, and P50 rules.

Table E.1 reports representative points from the three schedules. The full schedules contain one cutoff for every model age. Employment remains endogenous at age 75, as in the rest of the horizon, and the terminal cutoff is computed from positive baseline earnings at that age whenever the simulated sample contains them. If a finite simulation contains no positive terminal earnings, the implementation sets that cutoff to zero rather than inventing a positive threshold. The imposed-choice restriction is maintained through the terminal period in either case.

Table E.1: Age-specific earnings cutoffs for the income-contingent rules

Cutoff	Age 25	Age 40	Age 55
P10	£9,740	£10,168	£9,635
P25 (Q1)	£11,317	£13,616	£13,069
P50	£15,201	£18,936	£18,225

Notes: Annual 2015 pounds. Each cutoff is the corresponding percentile of positive labor earnings at that age in the baseline simulation.

All counterfactuals reported in the main figures resolve the decision problem. For the health-contingent rules, backward induction restricts the discrete choice set to the mandated choices in each named health state. For the earnings-targeted rules, if offered earnings w_{ij} are at or below the age-specific cutoff, only bundles with both healthy choices remain available, whatever the employment choice. In both cases the restriction enters the continuation value, so employment, consumption, and saving are optimal given the full future rule.

We also solved the earnings-targeted rules with eligibility based on realized earnings $n_{ij}w_{ij}$. Under that definition every nonworker is eligible, so 46.91%, 55.81%, and 70.65% of baseline person-years fall under the P10, P25, and P50 rules, and individuals whose offered earnings exceed the cutoff can leave the mandate by working. This escape response dominates the results. Under the P50 rule, mean cumulative earnings in the top decile of baseline cumulative earnings rise by 9.7%, almost as much as in the bottom decile; the variance of log cumulative earnings falls by only 4.96%, 1.28%, and 1.60% under the three rules. Its incidence therefore reflects the incentive to avoid the rule rather than the resources the rule requires, which is why the main text uses offered earnings. The price reduction instead modifies a model parameter and resolves the unrestricted choice problem at the new price.

Table E.3 decomposes the response to the 90% reduction in p_f . Panel A fixes groups using each individual's cumulative earnings in the paired baseline, so changes in group membership cannot generate the pattern mechanically. The decline in employment and cumulative earnings is largest below the median and becomes progressively smaller toward the top. Panel B shows the same outcomes by permanent health type. Type 2, the profile associated primarily with BB , has the lowest baseline cumulative earnings on average and responds most strongly. The decomposition therefore attributes the increase in overall dispersion to heterogeneous labor-supply responses concentrated toward the lower end of the baseline distribution, rather than to a contraction of earnings at the top.

Table E.2 decomposes the universal mandate and the P50 offered-earnings rule by baseline cumulative-earnings position. The gains are largest at the bottom of the baseline distribution, and under the P50 rule they are concentrated almost entirely below the median.

Table E.2: Distributional decomposition of the universal mandate and the P50 offered-earnings rule

Baseline group	N	Δ mean cumulative earnings (%)
<i>Panel A: Universal mandate</i>		
P0–P10	4,780	15.05
P10–P50	19,116	7.01
P50–P90	19,116	3.68
P90–P100	4,780	1.98
<i>Panel B: P50 offered-earnings rule</i>		
P0–P10	4,780	14.61
P10–P50	19,116	5.48
P50–P90	19,116	1.33
P90–P100	4,779	0.09

Notes: Individuals alive at age 55 in both simulations, grouped by baseline cumulative earnings over ages 25–55.

Table E.3: Distributional decomposition of the healthy-consumption price experiment

Baseline group	N	Δ work share (pp)	Δ mean cumulative earnings (%)
<i>Panel A: Baseline cumulative-earnings position</i>			
P0–P10	4,788	–1.27	–1.60
P10–P50	19,150	–1.32	–1.59
P50–P90	19,149	–0.92	–1.07
P90–P100	4,788	–0.48	–0.55
<i>Panel B: Permanent health type</i>			
Type 1 (<i>GG</i> profile)	22,970	–0.99	–1.00
Type 2 (<i>BB</i> profile)	2,308	–1.52	–2.49
Type 3 (<i>GB</i> profile)	7,054	–1.31	–1.49
Type 4 (<i>BG</i> profile)	15,543	–1.02	–1.12

Notes: The sample contains individuals alive at age 55 in both the baseline and counterfactual simulations. Panel A assigns individuals to groups using baseline cumulative earnings over ages 25–55. The work share is the fraction of alive person-years spent employed over the same ages. Panel B uses the permanent health-type labels defined in Section 3.2.1. Offered earnings schedules are unchanged by the experiment. Initial assets are common across individuals in the model, so the decomposition does not attribute the heterogeneous response to initial-wealth differences.

The results are neither welfare comparisons nor a ranking of cost-effective policies. The interventions have different coverage and do not include a common program budget, implementation cost, or compensation scheme. The income thresholds are held at their baseline

monetary values rather than recomputed from the counterfactual earnings distribution. This avoids making an individual’s cutoff depend on the simultaneous choices of the whole population and gives agents a known age-specific schedule. Because eligibility depends on offered earnings, the fraction eligible stays close to the percentile used to name the rule; it moves only as the intervention changes health, and hence offered earnings, and survival.

Channel shutdowns

The channel exercises of Figure 21 assign every health state the value the channel takes in GG , so that health no longer varies along that dimension. For the employment cost $\psi_j^n(H)$, this amounts to setting the health intercepts Δ_h^n and interactions $\psi_{n,m}^h$ to zero; λ_H and $s(j, H)$ are levels per state and take the GG entry. The alternative anchor, the health-share-weighted mean, holds the population average of the channel fixed and removes only its dispersion; it is available in the replication code and gives the same ordering of channels. Every subset is re-solved from scratch and simulated with the baseline seed, so shocks are paired at the individual level. The common-survivor rule, positive-earner criterion, and ages 25–55 earnings window are defined as in the choice experiments, but the comparison sample is constructed separately within each baseline–counterfactual pair. Against a baseline variance of 0.190, the percentage changes in the variance of log cumulative earnings are -4.69 for the wage gradient, -9.93 for the utility cost of working, -0.18 for mortality, -11.82 for the wage gradient and the utility cost together, -4.89 and -9.96 for each of them with mortality, and -11.83 for all three.

Standard errors are reported for the structural parameters but not for the counterfactual statistics, which are point estimates conditional on $\hat{\vartheta}$. Propagating parameter uncertainty through the counterfactual battery would require re-solving and re-simulating the model at draws from the asymptotic distribution of $\hat{\vartheta}$. Differences of a few tenths of a percentage point between scenarios should not be read as meaningful.